Hybrid Likelihood Methods in the Frequency Domain for Time Series Estimation

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Abstract

This thesis considers a Hybrid likelihood for frequency-domain inference in time series models, combining Whittle's parametric likelihood with frequency-domain empirical likelihood through a tuning parameter $\alpha \in [0, 1]$. An extensive Monte Carlo study explores the finite-sample behaviour of the Hybrid estimator under non-Gaussian innovations, conditional heteroskedasticity, additive outliers, and model misspecification. The results suggest that for $\alpha > 0$ the Hybrid estimator typically behaves very similarly to the Whittle estimator, a pattern that is consistent through all configurations. Hybridisation is observed to mitigate some of the instability of empirical likelihood in adverse finite-sample settings, while differences relative to Whittle remain small and confined to particularly challenging configurations.

Keywords: Whittle likelihood; empirical likelihood; hybrid likelihood; spectral inference; robustness; time series; frequency domain.

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1 Introduction

A powerful framework in modern inference is that of estimating equations. It is driven by the idea that a model can be characterized through a set of conditions that its must satisfy. One specifies functions $g(x, \boldsymbol{\theta})$ satisfying $\mathbb{E}_{F_{\boldsymbol{\theta}}}[g(X, \boldsymbol{\theta}_0)] = 0$ at the true parameter $\boldsymbol{\theta}_0$, where $X \sim F_{\boldsymbol{\theta}}$. This idea underlies a wide family of estimators. For instance, in the independent and identically distributed (i.i.d.) setting, the sample mean solves the estimating equation $E[X - \theta_0] = 0$. Another example would be the maximum likelihood estimator that comes from solving the derivative of the log-likelihood set to zero.

This notion was formalized by Huber's theory of M-estimators (Huber, 1964), which treats estimators as optimizers of sample criteria built from such equations. The M-estimation framework brought with it a unified way of understanding Fisher consistency, robustness and asymptotic behavior. From this foundation, the Generalized Methods of Moments (GMM) extended the idea further by allowing more estimating equations than unknown parameters, which is an overidentified setting. This allows to combine multiple moment restrictions and it was shown to give efficient estimators when moments were chosen properly (Hansen, 1982).

The flexibility of moment-based inference eventually led to extensions for dependent data. An example is the Generalized Estimating Equations (GEE) of Liang and Zeger (1986), which adapted the estimating-equation logic to longitudinal and correlated observations by incorporating a working correlation structure. GEE demonstrated that moment-based inference could be applied beyond the i.i.d. world, while retaining interpretability and robustness. These developments, M-estimation, GMM and GEE, collectively revealed how rich and adaptable the estimating-equation framework can be.

Parallel to these advances, researchers sought likelihood-based procedures that would keep the advantages of estimating equations without requiring a fully specified parametric model. This led to the development of the Empirical Likelihood (EL) method, originally

introduced by Owen (1988), which constructs a nonparametric likelihood by assigning probabilities to observations subject to moment constraints. EL preserves many features of classical likelihoods, including Wilks-type chi-square limits (i.e. the empirical likelihood ratio converges to a χ^2 distribution) and a self-studentizing property whereby no separate variance estimation is required. The higher-order accuracy and efficiency properties of EL and generalized empirical likelihoods were studied extensively by Newey and Smith (2004), who clarified their connection to optimal GMM.

Building on EL, a broader class of generalized empirical likelihoods (GEL) emerged, including the exponential tilting (ET) method of Kitamura and Stutzer (1997) and the exponentially tilted empirical likelihood (ETEL) of Schennach (2007). These methods were developed primarily in the i.i.d. or GMM context, where moment conditions are treated as approximately independent and where theoretical results on robustness and higher-order efficiency are most complete. Schennach's ETEL estimator is particularly relevant here: it shows how one can modify or combine existing estimating equations within the generalized empirical likelihood framework to obtain improved robustness and efficiency. Although GEL methods can be applied to dependent data through GMM-style moment conditions, their theoretical development does not extend to the frequency-domain setting used for stationary time series.

A natural question coming from this literature is whether one could combine likelihood-based methods to exploit different strengths simultaneously. This idea of hybrid likelihoods, i.e. convex combinations of parametric and empirical components, was explored in depth by Hjort et al. (2018) and further developed in Dæhlen and Hjort (2024). Hybrid likelihoods in the i.i.d. case have shown that carefully mixing two seemingly incompatible inference principles can give estimators that are both robust and efficient. These contributions have significantly expanded our understanding of how likelihoods can be blended, tuned and adapted to various statistical contexts.

While the i.i.d. setting now enjoys a well-developed theoretical and methodological foundation for hybrid and empirical likelihoods, the situation is quite different for time series. Because observations are dependent, the usual empirical likelihood structure breaks

down and constructing valid moment conditions requires care. A key breakthrough was the work of Monti (1997), who introduced the frequency-domain empirical likelihood (FDEL) for stationary time series. Monti observed that the periodogram ordinates are approximately independent at Fourier frequencies, enabling an empirical likelihood construction via spectral estimating equations. This resulted in a nonparametric likelihood ratio statistic with an asymptotic chi-square limit, mirroring the i.i.d. theory, but built on frequency-domain approximations rather than independence.

Monti's work, and later extensions by Nordman and Lahiri (2007), provided a way to carry empirical likelihood ideas into the time-series world. At the same time, frequency-domain parametric inference, centered around the Whittle likelihood introduced by Whittle (1953), remained the canonical choice for estimating spectral parameters under correct model specification. Whittle's method is well known for its asymptotic efficiency but is also sensitive to heavy tails, outliers or structural deviations from the assumed model. Thus, in the frequency domain, we find two complementary approaches: Whittle likelihood offers parametric efficiency, whereas FDEL offers nonparametric robustness.

More recently, extensions of GEL methods to the frequency domain have appeared. For example, Zhang et al. (2022) developed an exponential tilted likelihood for stationary time series, adapting the exponential tilting idea to spectral estimating equations and addressing feasibility issues of FDEL such as the convex hull problem. These contributions demonstrate that GEL principles can indeed be transported to the spectral setting. However, they remain within the empirical likelihood family and do not combine parametric and nonparametric likelihoods in the frequency domain. To date, no work has incorporated the parametric Whittle likelihood into such constructions, nor explored hybrid likelihoods blending Whittle and FDEL.

Together, these contributions suggest a broad principle: likelihoods and estimating-equation methods can be combined or modified to inherit complementary strengths. This idea is well developed in the i.i.d. case, but has not been explored in the frequency-domain context, where dependence complicates the construction of likelihoods and theoretical guarantees.

This observation provides the motivation and direction for the present work. Building on the spectral estimating-equation foundation shared by both Whittle likelihood and FDEL, we propose a Hybrid Likelihood that interpolates between the two through a tuning parameter $\alpha \in [0, 1]$. When $\alpha = 1$, the hybrid reduces to Whittle's method; when $\alpha = 0$, it becomes the FDEL. For intermediate values, it blends parametric efficiency with nonparametric robustness. In this sense, the hybrid acts as a frequency-domain analogue to the i.i.d. hybrid likelihoods of Dæhlen and Hjort (2024), but in an environment where the theoretical foundation is far less developed and the potential benefits for robust inference are substantial.

Conceptually, this thesis follows a *fil rouge* connecting three strands of modern inference: (i) estimating equations and M-estimation; (ii) empirical likelihood and its hybrids in the i.i.d. setting; and (iii) frequency-domain inference for stationary time series. By synthesizing these ideas, we aim to extend hybrid likelihood methodology into an area where it has not yet been explored, thereby offering a new pathway for robust and efficient spectral estimation.

2 Theoritical Background

This chapter collects the theoretical background that motivates the hybrid likelihood developed in this thesis. The material is not intended to introduce estimation procedures that are used directly in subsequent chapters, but rather to establish a common framework linking estimating equations, moment-based inference and likelihood-based methods. In particular, estimating equations, GMM and GEE are presented to clarify how parametric and nonparametric principles can be combined, and to highlight the conceptual foundations underlying empirical likelihood and hybrid constructions. Chapter 3 then specializes these ideas to the frequency domain, where Whittle likelihood, frequency-domain empirical likelihood and their hybrid counterpart are formally defined.

2.1 Estimating Equations and M-Estimation

Let $\{X_t\}_{t\geq 1}$ denote a stochastic process taking values in a measurable space $\mathcal{X}$. Assume $X_t \sim F_{\boldsymbol{\theta}}$, indexed by an unknown parameter vector $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^p$. Specify a measurable function $g : \mathcal{X} \times \Theta \to \mathbb{R}^q$, where q denotes the number of moment conditions, such that the true parameter $\boldsymbol{\theta}_0$ satisfies the population restriction

$$\mathbb{E}_{F_{\boldsymbol{\theta}}}\big[\, g(X_t, \boldsymbol{\theta}_0) \,\big] = \mathbf{0}. \tag{2.1}$$

When $q \geq p$ and the mapping $\boldsymbol{\theta} \mapsto \mathbb{E}_{F_{\boldsymbol{\theta}}}[g(X_t, \boldsymbol{\theta})]$ has a unique zero at $\boldsymbol{\theta}_0$, with full-rank Jacobian $\mathbb{E}[\nabla_{\boldsymbol{\theta}} g(X_t, \boldsymbol{\theta}_0)]$, the condition (2.1) identifies $\boldsymbol{\theta}_0$.

Given observations $\{X_t\}_{t=1}^n$, the empirical analogue of (2.1) is

$$G_n(\boldsymbol{\theta}) = \frac{1}{n}\sum_{t=1}^{n} g(X_t, \boldsymbol{\theta}), \tag{2.2}$$

and an estimator $\widehat{\boldsymbol{\theta}}$ satisfying

$$G_n(\widehat{\boldsymbol{\theta}}) = \mathbf{0} \tag{2.3}$$

is called a Z-estimator. Many standard estimators can be expressed in this form.

For instance, maximum likelihood estimation solves the score equation

$$\frac{1}{n}\sum_{t=1}^{n} \nabla_{\boldsymbol{\theta}} \log f(X_t; \boldsymbol{\theta}) = \mathbf{0}, \tag{2.4}$$

which corresponds to $g(x, \boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \log f(x; \boldsymbol{\theta})$. Empirical likelihood (EL), introduced by Owen (1988, 2001), provides a likelihood-based implementation of Z-estimation by assigning probabilities $\{p_t\}$ satisfying

$$\sum_{t=1}^{n} p_t\, g(X_t, \boldsymbol{\theta}) = \mathbf{0}, \tag{2.5}$$

and

$$\sum_{t=1}^{n} p_t = 1, \tag{2.6}$$

and maximizing the nonparametric likelihood

$$\prod_{t=1}^{n} np_t. \tag{2.7}$$

A closely related and more general formulation is that of M-estimation (Huber, 1964). Rather than enforcing the estimating equation directly, an M-estimator minimizes a sample criterion:

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\Theta} M_n(\boldsymbol{\theta}), \qquad M_n(\boldsymbol{\theta}) = \frac{1}{n}\sum_{t=1}^{n} m(X_t, \boldsymbol{\theta}), \tag{2.8}$$

for some function $m : \mathcal{X} \times \Theta \to \mathbb{R}$. When m is differentiable, the associated first-order condition is

$$\frac{1}{n}\sum_{t=1}^{n} \nabla_{\boldsymbol{\theta}} m(X_t, \boldsymbol{\theta}) = \mathbf{0}, \tag{2.9}$$

showing that M-estimators form a subclass of Z-estimators with $g(x, \boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} m(x, \boldsymbol{\theta})$.

Connection to hybrid likelihoods. In the i.i.d. setting, hybrid likelihoods combine parametric and empirical components by convex combination (Hjort et al., 2018; Dæhlen and Hjort, 2024):

$$\ell_{HL}(\boldsymbol{\theta}; \alpha) = \alpha\, \ell(\boldsymbol{\theta}) + (1-\alpha)\, \ell_{EL}(\boldsymbol{\theta}), \qquad \alpha \in [0, 1]. \tag{2.10}$$

This construction motivates the frequency-domain hybrid likelihood developed in Chapter 3, where Whittle likelihood and frequency-domain empirical likelihood are combined analogously.

Asymptotic behaviour of M- and Z-estimators. Under standard identification, smoothness, and moment conditions (Newey and McFadden, 1994; van der Vaart, 1998),

both Z-estimators and M-estimators are consistent for $\boldsymbol{\theta}_0$ and satisfy

$$\sqrt{n}\,(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0},\, \boldsymbol{V}), \tag{2.11}$$

where $\boldsymbol{V}$ denotes the asymptotic covariance matrix implied by the estimating function g or the criterion function m. This estimating-equation viewpoint provides the formal link between parametric likelihood, empirical likelihood, and hybrid constructions used in later chapters.

2.2 Generalized Method of Moments (GMM)

The estimating-equation framework extends naturally to the Generalized Method of Moments (GMM), which allows the number of moment conditions to exceed the number of parameters. Let $\boldsymbol{g}(x, \boldsymbol{\theta}) \in \mathbb{R}^q$ denote a vector of moment functions such that

$$\mathbb{E}_{F_{\boldsymbol{\theta}}}\big[\, \boldsymbol{g}(X_t, \boldsymbol{\theta}_0) \,\big] = \mathbf{0}, \qquad q \geq p, \tag{2.12}$$

where $\boldsymbol{\theta}_0 \in \mathbb{R}^p$ is the true parameter vector. When $q > p$, the system is overidentified and the moment conditions cannot generally be enforced exactly at the sample level.

Given observations $\{X_t\}_{t=1}^n$, define the sample moments

$$\boldsymbol{G}_n(\boldsymbol{\theta}) = \frac{1}{n}\sum_{t=1}^{n} \boldsymbol{g}(X_t, \boldsymbol{\theta}) \;\in\; \mathbb{R}^q. \tag{2.13}$$

The GMM estimator is defined as

$$\widehat{\boldsymbol{\theta}}_{GMM} = \arg\min_{\boldsymbol{\theta}\in\Theta} Q_n(\boldsymbol{\theta}), \qquad Q_n(\boldsymbol{\theta}) = \boldsymbol{G}_n(\boldsymbol{\theta})^\top \boldsymbol{W}_n \boldsymbol{G}_n(\boldsymbol{\theta}), \tag{2.14}$$

where $\boldsymbol{W}_n$ is a positive definite weighting matrix. When $q = p$ and $\boldsymbol{W}_n$ is fixed, minimizing (2.14) recovers the Z-estimator of Section 2.1.

Let

$$\boldsymbol{S} = \mathrm{Var}\big(\sqrt{n}\,\boldsymbol{G}_n(\boldsymbol{\theta}_0)\big)\,, \qquad \boldsymbol{D} = \mathbb{E}[\nabla_{\boldsymbol{\theta}} \boldsymbol{g}(X_t, \boldsymbol{\theta}_0)] \in \mathbb{R}^{q\times p}, \tag{2.15}$$

and assume that $\boldsymbol{D}$ has full column rank. Under standard regularity conditions, including identification of $\boldsymbol{\theta}_0$, smoothness of the moment functions $\boldsymbol{g}(x, \boldsymbol{\theta})$, finite second moments, and a law of large numbers and central limit theorem for $\boldsymbol{G}_n(\boldsymbol{\theta}_0)$ (Hansen, 1982; Newey and McFadden, 1994),

the GMM estimator satisfies

$$\sqrt{n}\,(\widehat{\boldsymbol{\theta}}_{GMM} - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}\big(\mathbf{0},\ (\boldsymbol{D}^\top \boldsymbol{W} \boldsymbol{D})^{-1} \boldsymbol{D}^\top \boldsymbol{W} \boldsymbol{S} \boldsymbol{W} \boldsymbol{D} (\boldsymbol{D}^\top \boldsymbol{W} \boldsymbol{D})^{-1}\big)\,, \tag{2.16}$$

where $\boldsymbol{W}$ denotes the probability limit of $\boldsymbol{W}_n$. Efficiency is achieved by choosing

$$\boldsymbol{W} = \boldsymbol{S}^{-1}, \tag{2.17}$$

which yields asymptotic covariance

$$(\boldsymbol{D}^\top \boldsymbol{S}^{-1} \boldsymbol{D})^{-1}. \tag{2.18}$$

Relation to empirical likelihood and GEL. Maximizing the empirical likelihood subject to the moment constraint

$$\sum_{t=1}^{n} p_t \boldsymbol{g}(X_t, \boldsymbol{\theta}) = \mathbf{0} \tag{2.19}$$

leads to an estimator that is asymptotically equivalent to efficient GMM (Owen, 1988; Newey and Smith, 2004). Generalized empirical likelihood (GEL) methods, such as exponential tilting (Kitamura and Stutzer, 1997) and exponentially tilted empirical likelihood (ETEL) (Schennach, 2007), modify how the same moment conditions are imposed while retaining this asymptotic equivalence. Although these methods can be extended to dependent data through time-domain moment restrictions, their theory does not directly apply to the frequency-domain setting considered here.

2.3 Generalized Estimating Equations (GEE)

The estimating-equation framework developed in the previous sections applies naturally to independent observations. Many applications, however, involve dependent or correlated data structures for which independence-based methods are no longer appropriate. Generalized Estimating Equations (GEE), introduced by Liang and Zeger (1986), extend estimating-equation methodology to such settings by explicitly accounting for within-cluster dependence.

GEE requires specification of only the first two moments of the data and avoids the need for a full probabilistic model for the joint distribution of the process. Let $\boldsymbol{\mu}_t(\boldsymbol{\theta}) = \mathbb{E}[X_t \mid \boldsymbol{\theta}]$ denote the conditional mean, and define the residuals

$$\boldsymbol{r}_t(\boldsymbol{\theta}) = X_t - \boldsymbol{\mu}_t(\boldsymbol{\theta}). \tag{2.20}$$

Assume observations are grouped into clusters within which dependence is present. A working correlation structure $\boldsymbol{R}(\alpha)$ is introduced to model within-cluster dependence. The corresponding working covariance matrix is

$$\boldsymbol{W}_t(\boldsymbol{\theta}, \alpha) = \boldsymbol{A}_t(\boldsymbol{\theta})^{1/2} \boldsymbol{R}(\alpha) \boldsymbol{A}_t(\boldsymbol{\theta})^{1/2}, \tag{2.21}$$

where $\boldsymbol{A}_t(\boldsymbol{\theta})$ contains the marginal variances.

The generalized estimating equation for $\boldsymbol{\theta}$ is

$$\sum_{t=1}^{T} \boldsymbol{D}_t(\boldsymbol{\theta})^\top \boldsymbol{W}_t(\boldsymbol{\theta}, \alpha)^{-1} \boldsymbol{r}_t(\boldsymbol{\theta}) = \boldsymbol{0}, \tag{2.22}$$

where

$$\boldsymbol{D}_t(\boldsymbol{\theta}) = \frac{\partial \boldsymbol{\mu}_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}^\top}. \tag{2.23}$$

Equation (2.22) generalizes the Z-estimation condition $G_n(\boldsymbol{\theta}) = \boldsymbol{0}$ to correlated data by weighting the residuals through the working covariance $\boldsymbol{W}_t$.

A key feature of GEE is that the working correlation $\boldsymbol{R}(\alpha)$ need not be correctly

specified. Provided the mean model $\boldsymbol{\mu}_t(\boldsymbol{\theta})$ is correct, the solution $\widehat{\boldsymbol{\theta}}_{GEE}$ is consistent for $\boldsymbol{\theta}_0$, with misspecification of the covariance affecting only efficiency (Liang and Zeger, 1986). The asymptotic covariance matrix has the sandwich form

$$\boldsymbol{V}_{GEE} = \boldsymbol{A}^{-1}\boldsymbol{B}\boldsymbol{A}^{-1}, \tag{2.24}$$

where

$$\boldsymbol{A} = \sum_{t=1}^{T} \boldsymbol{D}_t(\boldsymbol{\theta}_0)^\top \boldsymbol{W}_t(\boldsymbol{\theta}_0, \alpha_0)^{-1} \boldsymbol{D}_t(\boldsymbol{\theta}_0), \tag{2.25}$$

and

$$\boldsymbol{B} = \sum_{t=1}^{T} \boldsymbol{D}_t(\boldsymbol{\theta}_0)^\top \boldsymbol{W}_t(\boldsymbol{\theta}_0, \alpha_0)^{-1} \mathrm{Cov}(X_t)\, \boldsymbol{W}_t(\boldsymbol{\theta}_0, \alpha_0)^{-1} \boldsymbol{D}_t(\boldsymbol{\theta}_0). \tag{2.26}$$

Connection to spectral inference. GEE is included to emphasize that estimating-equation methods remain valid in the presence of dependence when appropriate weighting is used. This principle underlies the frequency-domain approach adopted later, where dependence is handled through spectral transformations rather than working covariance structures.

2.4 Spectral Density and the Periodogram

Time series consist of observations recorded over time, and their statistical analysis is distinguished by the presence of serial dependence. Formally, consider a stochastic process $\{X_t,\, t \in \mathbb{Z}\}$. Under stationarity, a broad and flexible class of models admits the causal representation

$$X_t = G(\ldots, Z_{t-1}, Z_t; \boldsymbol{\theta}_0), \tag{2.27}$$

where G is a measurable function depending on an unknown parameter vector $\boldsymbol{\theta}_0 = (\theta_0^{(1)}, \ldots, \theta_0^{(p)})$ and $\{Z_t\}$ is a sequence of i.i.d. innovations. This formulation encompasses many linear and nonlinear processes, including ARMA models (Box and Jenkins, 1970, Chapter 3) and nonlinear causal processes (Shao and Wu, 2007).

Time-domain methods based on autocovariances and autocorrelations are commonly

used to study such processes. However, when strong cyclical components, hidden periodicities, or long-range dependence are present, frequency-domain techniques often provide a more informative description. The key idea is to decompose the variability of the process into contributions from different frequencies (Priestley, 1981; Brillinger, 1981). An additional advantage is computational: the Fourier transform approximately diagonalizes the dependence structure, simplifying likelihood-based procedures relative to time-domain approaches that rely on large covariance matrices (Brockwell and Davis, 1991). These considerations motivate the introduction of the spectral density and the periodogram.

Spectral Density. Let (X_t) be weakly stationary with mean zero and autocovariance function

$$\gamma_X(h) = \mathbb{E}[X_0 X_h]. \tag{2.28}$$

Under the short-range dependence condition

$$\sum_{h=-\infty}^{\infty} |\gamma_X(h)| < \infty, \tag{2.29}$$

the process admits a spectral representation (Priestley, 1981; Brillinger, 1981), and the spectral density is given by the Fourier transform of the autocovariance function:

$$f(\lambda) = \frac{1}{2\pi} \sum_{h \in \mathbb{Z}} \gamma_X(h)\, e^{-i\lambda h}, \qquad \lambda \in [-\pi, \pi]. \tag{2.30}$$

The function $f(\lambda)$ describes how the variance of the process is distributed across frequencies and plays a central role in frequency-domain inference.

Discrete Fourier Transform and the Periodogram. Given observations $X_1, \ldots, X_n$, the frequency-domain analysis is based on the discrete Fourier transform (DFT) evaluated at the Fourier frequencies

$$\lambda_{j,n} = \frac{2\pi j}{n}, \qquad j \in I_n := \left\{ -\left\lfloor \frac{n-1}{2} \right\rfloor, \ldots, \left\lfloor \frac{n-1}{2} \right\rfloor \right\}. \tag{2.31}$$

The DFT is defined by

$$d_n^X(\lambda_{j,n}) = \sum_{t=1}^{n} X_t \, e^{i\lambda_{j,n} t}, \tag{2.32}$$

and the associated periodogram is

$$I_n^X(\lambda_{j,n}) = \frac{1}{2\pi n} \left| d_n^X(\lambda_{j,n}) \right|^2. \tag{2.33}$$

Asymptotic Distribution of Periodogram Ordinates

For weakly dependent stationary processes, the DFT coefficients at distinct Fourier frequencies are asymptotically uncorrelated and jointly complex normal as $n \to \infty$ (Brillinger, 1981). This result extends to general nonlinear causal processes (Shao and Wu, 2007). Consequently, the corresponding periodogram ordinates behave asymptotically as independent exponential random variables scaled by the true spectral density.

More precisely, for any fixed distinct frequencies $\lambda^{(1)}, \ldots, \lambda^{(k)} \in (0, \pi)$,

$$\sup_{x_1,\ldots,x_k \geq 0} \left| P\big(I_n^X(\lambda^{(1)}) \leq x_1, \ldots, I_n^X(\lambda^{(k)}) \leq x_k\big) - P\left(\frac{W_1}{f(\lambda^{(1)})} \leq x_1, \ldots, \frac{W_k}{f(\lambda^{(k)})} \leq x_k \right) \right| \longrightarrow 0, \tag{2.34}$$

as $n \to \infty$, where $W_1, \ldots, W_k$ are independent Exp(1) random variables. Equivalently, for each fixed $\lambda \in (0, \pi)$,

$$\frac{I_n^X(\lambda)}{f(\lambda)} \xrightarrow{d} \mathrm{Exp}(1). \tag{2.35}$$

Thus, in large samples,

$$I_n^X(\lambda) \approx f(\lambda)\, W, \qquad W \sim \mathrm{Exp}(1). \tag{2.36}$$

The approximate independence and exponential behaviour of the periodogram ordinates provide the basis for likelihood- and moment-based inference procedures in the

frequency domain, developed in Chapter 3.

3 Whittle Likelihood, Frequency-Domain Empirical Likelihood, and the Hybrid Method

This chapter introduces the frequency-domain likelihoods that form the building blocks of the proposed hybrid method. We first review the Whittle likelihood and the frequency-domain empirical likelihood (FDEL), fixing notation and highlighting their common spectral estimating equations. We then use this shared structure to construct a hybrid likelihood that interpolates between the two approaches.

3.1 Whittle Likelihood

Let $\{X_t\}_{t=1}^n$ be a zero-mean, weakly stationary time series with spectral density $f(\lambda;\boldsymbol{\theta})$ indexed by a parameter vector $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^p$. Building on the frequency-domain approximation developed in Section 2.4, Whittle (Whittle, 1953) proposed a likelihood-based estimator obtained by treating the periodogram ordinates as approximately independent with mean $f(\lambda;\boldsymbol{\theta})$.

Under this approximation, the negative log-likelihood in the frequency domain is, up to an additive constant,

$$\ell_{WL}(\boldsymbol{\theta}) = -\sum_{\lambda_j \in \Lambda_n} \left[\log f(\lambda_j;\boldsymbol{\theta}) + \frac{I(\lambda_j)}{f(\lambda_j;\boldsymbol{\theta})} \right], \tag{3.1}$$

which is known as the Whittle likelihood. The Whittle estimator is defined as the maximizer

$$\widehat{\boldsymbol{\theta}}_{WL} = \arg\max_{\boldsymbol{\theta}\in\Theta} \ell_{WL}(\boldsymbol{\theta}). \tag{3.2}$$

Whittle score and estimating equations

Differentiating (3.1) with respect to $\boldsymbol{\theta}$ yields the frequency-domain score contributions

$$\boldsymbol{\psi}_j(\boldsymbol{\theta}) = \left(\frac{I(\lambda_j)}{f(\lambda_j; \boldsymbol{\theta})} - 1 \right) \nabla_{\boldsymbol{\theta}} \log f(\lambda_j; \boldsymbol{\theta}), \tag{3.3}$$

so that the Whittle estimator solves the spectral estimating equation

$$\sum_{\lambda_j \in \Lambda_n} \boldsymbol{\psi}_j(\boldsymbol{\theta}) = \mathbf{0}. \tag{3.4}$$

This representation makes explicit that Whittle likelihood is an M-estimation procedure in the frequency domain, based on the same spectral estimating functions that will be used in the empirical likelihood and hybrid constructions developed below.

3.2 Frequency-Domain Empirical Likelihood (FDEL)

The empirical likelihood (EL) approach of Owen (1988) can be adapted to the frequency domain by treating the spectral estimating equations $\{\boldsymbol{\psi}_j(\boldsymbol{\theta})\}_{\lambda_j \in \Lambda_n}$ as moment conditions. This idea was developed by Monti (1997), who constructed the frequency-domain empirical likelihood (FDEL) exploiting the approximate independence of periodogram ordinates at Fourier frequencies.

Empirical likelihood weights

Let the spectral moment conditions satisfy

$$\mathbb{E}\big[\boldsymbol{\psi}_j(\boldsymbol{\theta}_0)\big] = \mathbf{0}, \qquad \lambda_j \in \Lambda_n. \tag{3.5}$$

FDEL assigns probability weights $\{p_j(\boldsymbol{\theta})\}_{\lambda_j \in \Lambda_n}$ to the frequencies $\Lambda_n = \{\lambda_1, \ldots, \lambda_m\}$ and maximizes the empirical likelihood

$$L_{EL}(\boldsymbol{\theta}) = \max_{\{p_j\}} \left\{ \prod_{\lambda_j \in \Lambda_n} m\, p_j(\boldsymbol{\theta}) \;:\; \sum_{\lambda_j \in \Lambda_n} p_j(\boldsymbol{\theta}) \boldsymbol{\psi}_j(\boldsymbol{\theta}) = \mathbf{0},\; \sum_{\lambda_j \in \Lambda_n} p_j(\boldsymbol{\theta}) = 1 \right\}. \tag{3.6}$$

As in the i.i.d. empirical likelihood, the maximizing weights admit the explicit form

$$p_j(\boldsymbol{\theta}) = \frac{1}{m} \frac{1}{1 + \boldsymbol{\xi}(\boldsymbol{\theta})^\top \boldsymbol{\psi}_j(\boldsymbol{\theta})}, \qquad \lambda_j \in \Lambda_n, \tag{3.7}$$

where $\boldsymbol{\xi}(\boldsymbol{\theta}) \in \mathbb{R}^p$ is the Lagrange multiplier associated with the moment constraints. It is determined as the solution of

$$\sum_{\lambda_j \in \Lambda_n} \frac{\boldsymbol{\psi}_j(\boldsymbol{\theta})}{1 + \boldsymbol{\xi}(\boldsymbol{\theta})^\top \boldsymbol{\psi}_j(\boldsymbol{\theta})} = \mathbf{0}. \tag{3.8}$$

Empirical log-likelihood

Substituting the weights (3.7) into (3.6) yields the frequency-domain empirical log-likelihood

$$\ell_{EL}(\boldsymbol{\theta}) = -\sum_{\lambda_j \in \Lambda_n} \log\big(1 + \boldsymbol{\xi}(\boldsymbol{\theta})^\top \boldsymbol{\psi}_j(\boldsymbol{\theta})\big)\,, \tag{3.9}$$

The FDEL estimator is then defined as

$$\widehat{\boldsymbol{\theta}}_{EL} = \arg\max_{\boldsymbol{\theta} \in \Theta} \ell_{EL}(\boldsymbol{\theta}). \tag{3.10}$$

Asymptotic properties

Under standard regularity conditions, including short-range dependence of the underlying process and smoothness and identifiability of the spectral density, Monti (1997) showed that the empirical likelihood ratio statistic

$$R_{EL}(\boldsymbol{\theta}_0) = 2\Big[\ell_{EL}(\widehat{\boldsymbol{\theta}}_{EL}) - \ell_{EL}(\boldsymbol{\theta}_0)\Big] \tag{3.11}$$

satisfies the Wilks-type limit

$$R_{EL}(\boldsymbol{\theta}_0) \xrightarrow{d} \chi^2_p. \tag{3.12}$$

This result parallels classical empirical likelihood theory and provides a nonparametric likelihood-based inference procedure without assuming a correctly specified parametric spectral model.

Extensions to short-memory, long-memory, and nonlinear processes were developed by Nordman (2004) and Nordman and Lahiri (2007), establishing consistency and chi-square limits under broad dependence structures.

Relation to Whittle likelihood and generalized EL

FDEL and Whittle likelihood rely on the same spectral estimating equations $\{\boldsymbol{\psi}_j(\boldsymbol{\theta})\}$ but differ fundamentally in how these equations are imposed: Whittle likelihood incorporates them parametrically through an M-estimation criterion. FDEL enforces them exactly as empirical moment constraints in a Z-estimation framework. They therefore represent complementary likelihood-based approaches built from the same frequency-domain information.

3.3 Hybrid Likelihood Construction

The Whittle likelihood (Section 3.1) and the frequency-domain empirical likelihood (Section 3.2) are both based on the same spectral estimating functions

$$\boldsymbol{\psi}_j(\boldsymbol{\theta}) = \left(\frac{I(\lambda_j)}{f(\lambda_j; \boldsymbol{\theta})} - 1 \right) \nabla_{\boldsymbol{\theta}} \log f(\lambda_j; \boldsymbol{\theta}), \tag{3.13}$$

but impose these conditions through distinct inferential mechanisms. This shared structure provides a natural basis for combining the two approaches within a single likelihood framework. Motivated by the hybrid likelihood constructions developed in the i.i.d. setting by Hjort et al. (2018) and Dæhlen and Hjort (2024), we construct a frequency-domain hybrid likelihood by blending the Whittle and FDEL criteria through a convex combina-

tion.

Definition

For a tuning parameter $\alpha \in [0,1]$, the Hybrid Likelihood (HL) is defined as

$$\ell_{HL}(\boldsymbol{\theta};\alpha) = \alpha\,\ell_{WL}(\boldsymbol{\theta}) + (1-\alpha)\,\ell_{EL}(\boldsymbol{\theta}), \tag{3.14}$$

where ℓ_{WL} and ℓ_{EL} are given in (3.1) and (3.9), respectively. The corresponding hybrid estimator is

$$\widehat{\boldsymbol{\theta}}_{HL}(\alpha) = \arg\max_{\boldsymbol{\theta}\in\Theta} \ell_{HL}(\boldsymbol{\theta};\alpha). \tag{3.15}$$

The extreme cases recover the benchmark estimators:

$$\alpha = 1 \;\Rightarrow\; \widehat{\boldsymbol{\theta}}_{HL} = \widehat{\boldsymbol{\theta}}_{WL}, \qquad \alpha = 0 \;\Rightarrow\; \widehat{\boldsymbol{\theta}}_{HL} = \widehat{\boldsymbol{\theta}}_{EL}. \tag{3.16}$$

Thus, the hybrid estimator interpolates continuously between the parametric Whittle estimator and the nonparametric FDEL estimator.

Hybrid estimating equations

Differentiating (3.14) with respect to $\boldsymbol{\theta}$ yields the hybrid estimating equation

$$\alpha \sum_{\lambda_j\in\Lambda_n} \boldsymbol{\psi}_j(\boldsymbol{\theta}) \;-\; (1-\alpha) \sum_{\lambda_j\in\Lambda_n} \frac{\boldsymbol{\psi}_j(\boldsymbol{\theta})}{1+\boldsymbol{\xi}(\boldsymbol{\theta})^\top \boldsymbol{\psi}_j(\boldsymbol{\theta})} = \mathbf{0}, \tag{3.17}$$

where $\boldsymbol{\xi}(\boldsymbol{\theta})$ denotes the FDEL Lagrange multiplier. Equation (3.17) shows that the hybrid likelihood combines the parametric score contribution of the Whittle likelihood with the empirical likelihood score induced by the FDEL weights. Accordingly, the hybrid estimator is an M-estimator whose objective incorporates Z-estimation components, mirroring the structure of hybrid likelihood estimators in the i.i.d. setting.

Interpretation of α

The tuning parameter α determines the relative contribution of the parametric and empirical components in the hybrid likelihood. Values of α close to one place greater weight on the Whittle likelihood, while values close to zero emphasise the empirical likelihood component. Accordingly, α may be viewed as indexing a continuum between a fully parametric and a fully empirical frequency-domain likelihood.

The hybrid construction is motivated by the idea of balancing the potential efficiency of the Whittle likelihood under correct model specification with the greater robustness of empirical likelihood methods to model misspecification. This interpretation is heuristic: at this stage, no theoretical optimality or robustness properties are claimed for any particular choice of α. Instead, α is treated as a tuning parameter whose practical impact must be assessed empirically.

In Chapter 4, we therefore investigate the behaviour of $\widehat{\boldsymbol{\theta}}_{HL}(\alpha)$ for several fixed values of α across a range of data-generating processes, without presupposing a priori dominance of either the parametric or empirical component.

Relation to existing literature

The hybrid likelihood (3.14) is a frequency-domain analogue of convex hybrid likelihoods studied in the i.i.d. setting by Hjort et al. (2018) and Dæhlen and Hjort (2024). Unlike frequency-domain generalized empirical likelihood methods, such as the exponential tilting approach of Zhang et al. (2022), the proposed construction combines a parametric likelihood component with an empirical likelihood component.

4 Simulation Studies

In this section, we conduct Monte Carlo simulations to investigate the finite-sample performance of the proposed frequency-domain estimators under a variety of data-generating processes. The simulation design aims to assess both estimation accuracy and robustness

to departures from standard modelling assumptions.

We consider a range of linear and nonlinear time series models, including AR(1), ARMA(1,1), and ARFIMA$(1, d, 0)$ processes with $d \in \{0.1, 0.4\}$, as well as ARCH(1) models, AR(1) processes with additive outliers, and AR(1) models with skewed-t innovations. To study the impact of model misspecification, we additionally generate data from an AR(2) process while fitting an AR(1) model.

For each correctly specified model, except for the skewed-t case, four innovation distributions are considered: standard normal $N(0, 1)$, Student-t with five degrees of freedom, chi-square $\chi^2(5)$ innovations centered to have mean zero, and exponential Exp(1) innovations centered to have mean zero. The skewed-t innovation model is treated separately due to its inherent asymmetry. Sample sizes of $T = 50, 75, 100$ and 200 are examined, and each scenario is replicated 1000 times.

For correctly specified models, estimator performance is evaluated using bias, standard deviation (SD), and root mean squared error (RMSE). For a generic scalar parameter θ, the RMSE is defined as

$$\mathrm{RMSE}(\hat{\theta}) = \left\{ \frac{1}{R} \sum_{r=1}^{R} \left(\hat{\theta}^{(r)} - \theta_0 \right)^2 \right\}^{1/2}, \tag{4.1}$$

where $\hat{\theta}^{(r)}$ denotes the estimate obtained in replication r and θ_0 is the true parameter value. This criterion provides a joint measure of bias and variability and serves as the primary performance metric in the simulation study.

In the case of model misspecification, the true parameter value θ_0 is no longer well defined. Estimator performance is therefore assessed using a spectral distance measure that quantifies discrepancies between the fitted and true spectral densities. The precise definition of this distance, based on a trapezoidal approximation, is introduced later in the misspecification subsection.

For the Hybrid estimator, the tuning parameter α is selected in an oracle fashion. Specifically, for each simulation scenario, α is chosen from a fixed grid to minimize the Monte Carlo RMSE computed using knowledge of the true data-generating process. This

procedure does not correspond to a feasible data-driven tuning rule but instead provides an upper bound on the potential gains achievable by the Hybrid estimator relative to Whittle and FDEL.

Finally, we focus on the most informative configurations in which differences among the Whittle, FDEL, and Hybrid estimators are most pronounced. Results for additional innovation distributions and sample sizes exhibit qualitatively similar behaviour and are therefore reported in Appendix 9.

4.1 AR(1)

We begin by examining the finite-sample performance of the competing estimators under a correctly specified first-order autoregressive model. The data-generating process is given by

$$X_t = \phi_0 X_{t-1} + \varepsilon_t, \qquad t = 1, \ldots, T, \tag{4.2}$$

where $\phi_0 = 0.4$ and $\{\varepsilon_t\}$ denotes an i.i.d. innovation sequence with zero mean and finite variance.

Table 4.1: AR(1), $\phi_0 = 0.4$, t_5 innovations

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.02339	0.10839	-0.02341
	SD	0.13988	0.29105	0.13981
	RMSE	0.14176	0.31044	**0.14169**
				$\alpha^* = 0.05$
75	Bias	-0.00995	0.09323	-0.00995
	SD	0.10337	0.24797	0.10337
	RMSE	0.10380	0.26480	0.10380
				$\alpha^* = 0.35$
100	Bias	-0.01393	0.07094	-0.01393
	SD	0.09257	0.22948	0.09257
	RMSE	0.09356	0.24008	0.09356
				$\alpha^* = 0.50$
200	Bias	-0.00923	0.02824	-0.00923
	SD	0.06545	0.15856	0.06545
	RMSE	0.06607	0.16098	0.06607
				$\alpha^* = 0.10$

Table 4.2: AR(1), $\phi_0 = 0.4$, Exp(1) innovations

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.02350	0.10639	-0.02402
	SD	0.13896	0.30210	0.13742
	RMSE	0.14087	0.32014	**0.13944**
				$\alpha^* = 0.35$
75	Bias	-0.01562	0.09092	-0.01562
	SD	0.09952	0.24963	0.09952
	RMSE	0.10069	0.26555	0.10069
				$\alpha^* = 0.80$
100	Bias	-0.01395	0.06982	-0.01395
	SD	0.08926	0.22546	0.08926
	RMSE	0.09030	0.23591	0.09030
				$\alpha^* = 0.65$
200	Bias	-0.00143	0.05161	-0.00143
	SD	0.06389	0.17829	0.06389
	RMSE	0.06388	0.18552	0.06388
				$\alpha^* = 0.40$

Bold values indicate the smallest RMSE among the competing estimators. **Blue** denotes the Hybrid estimator, **green** the FDEL estimator, and **purple** the Whittle estimator.

Discussion

The small-sample results ($T = 50$) are the only settings in which visible differences arise between the three estimators in Tables 4.1 and 4.2. The remaining results are reported in Appendix 9, on Tables 9.1 and 9.2. Under heavy-tailed t_5 innovations and skewed Exp(1) innovations, the FDEL estimator exhibits substantially inflated variability, leading to larger RMSE values. By contrast, the Whittle estimator remains numerically stable even in very small samples.

The behaviour of the Hybrid estimator is particularly informative. Although it is defined as a convex combination of the Whittle and FDEL contrasts, its RMSE coincides with that of the Whittle estimator for all considered sample sizes. For $T = 50$, the

Hybrid estimator even attains a marginally smaller RMSE than Whittle, although these differences are numerically negligible.

These results indicate that the Hybrid estimator inherits the efficiency of the Whittle estimator as soon as the tuning parameter α is moved away from zero. The RMSE drops sharply between $\alpha = 0$ (pure FDEL) and $\alpha = 0.05$, as shown in Figure 4.2, while for $\alpha \geq 0.05$ the RMSE curve becomes essentially flat. As a consequence, the oracle tuning parameter α^* does not follow a smooth or interpretable pattern across simulation settings. Different values of α yield virtually identical estimators, and the observed variability in α^* therefore reflects numerical indifference rather than meaningful statistical trade-offs.

Behaviour of $\widehat{\phi}(\alpha)$ and error metrics across α

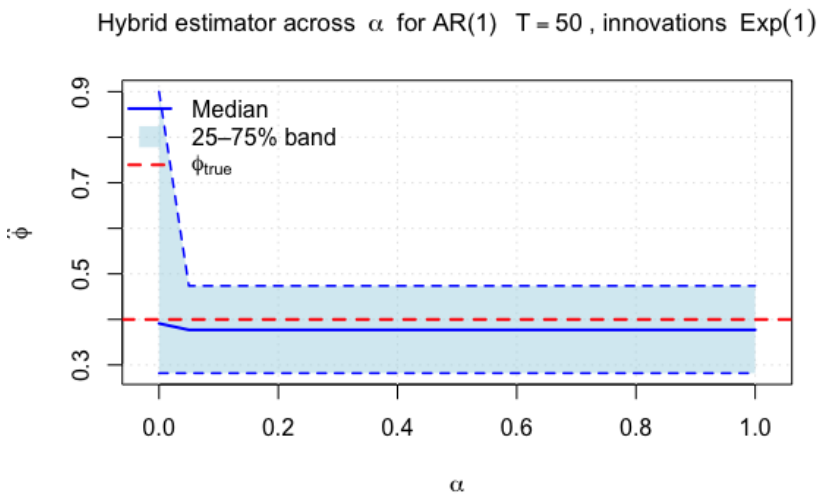


Figure 4.1: Functional boxplot of the hybrid estimator $\widehat{\phi}(\alpha)$ for the AR(1) model with $T = 50$ and Exp(1) innovations. The solid line represents the pointwise median of $\widehat{\phi}(\alpha)$, the shaded region denotes the interquartile (25–75%) band, and the dashed horizontal line indicates the true parameter value $\phi_0 = 0.4$. The figure illustrates the sharp transition from unstable FDEL behaviour at $\alpha = 0$ to a flat and stable regime once a positive Whittle weight is included.

Figure 4.1 illustrates the behaviour of the Hybrid estimator as a function of the tuning parameter for the representative case $T = 50$ under Exp(1) innovations. A sharp transition occurs immediately after $\alpha = 0$. The FDEL estimator is unstable in small samples, consistent with its reliance on moment conditions and approximate independence

properties that are only asymptotically valid. This leads to poorly behaved Lagrange multipliers and inflated variability.

Once α becomes positive, the Hybrid estimator collapses onto a flat plateau that coincides with the Whittle estimator. Both the median curve and the interquartile band stabilise immediately, showing that even a very small Whittle component suffices to regularise the estimator.

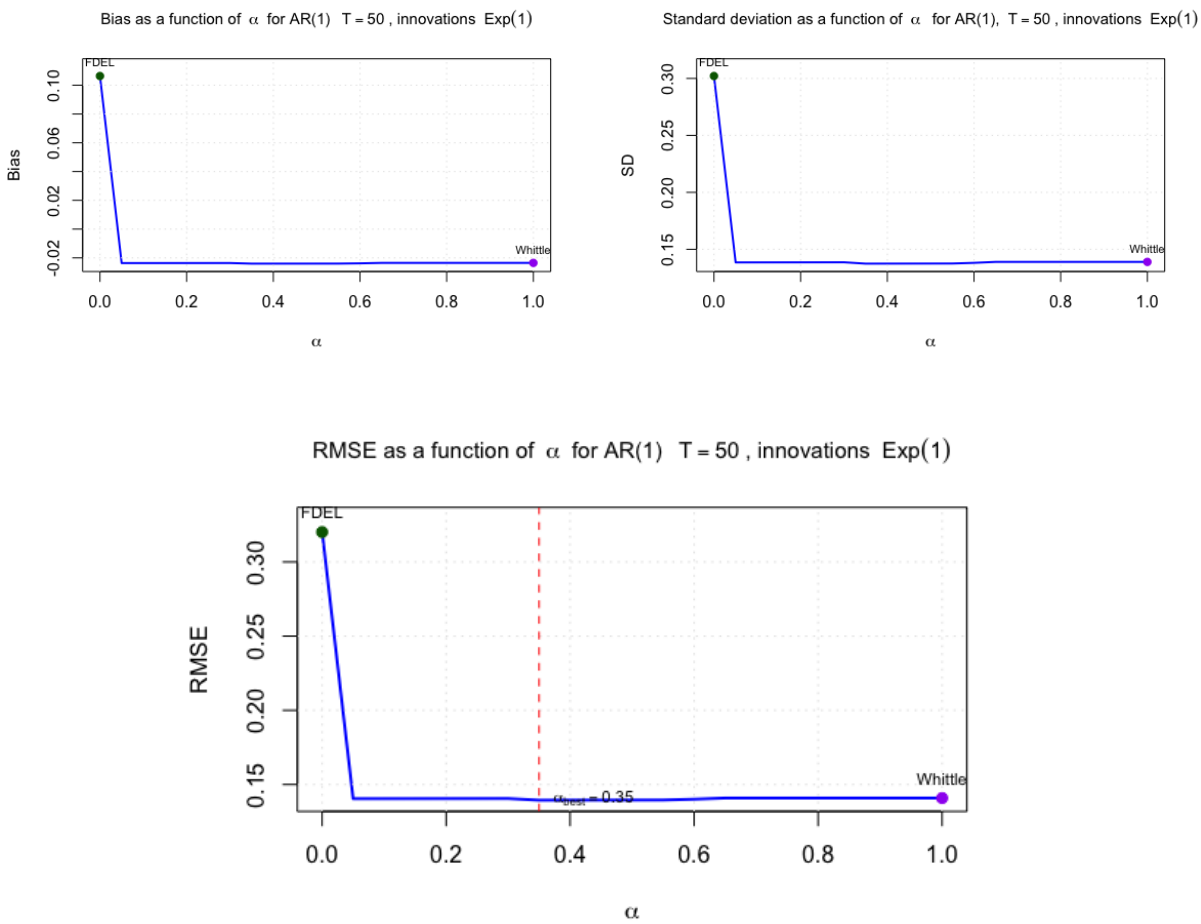


Figure 4.2: Bias, standard deviation, and RMSE as functions of α for AR(1), $T = 50$, Exp(1) innovations.

Figure 4.2 shows how bias, standard deviation, and RMSE vary with α. All three metrics display the same qualitative pattern: a sharp improvement when moving from $\alpha = 0$ to small positive values, followed by a long flat region. In particular, RMSE decreases abruptly for $\alpha \leq 0.05$ and remains essentially constant thereafter. The numerically selected value α^* lies well within this flat region, where RMSE differences are smaller

than 10^{-3}.

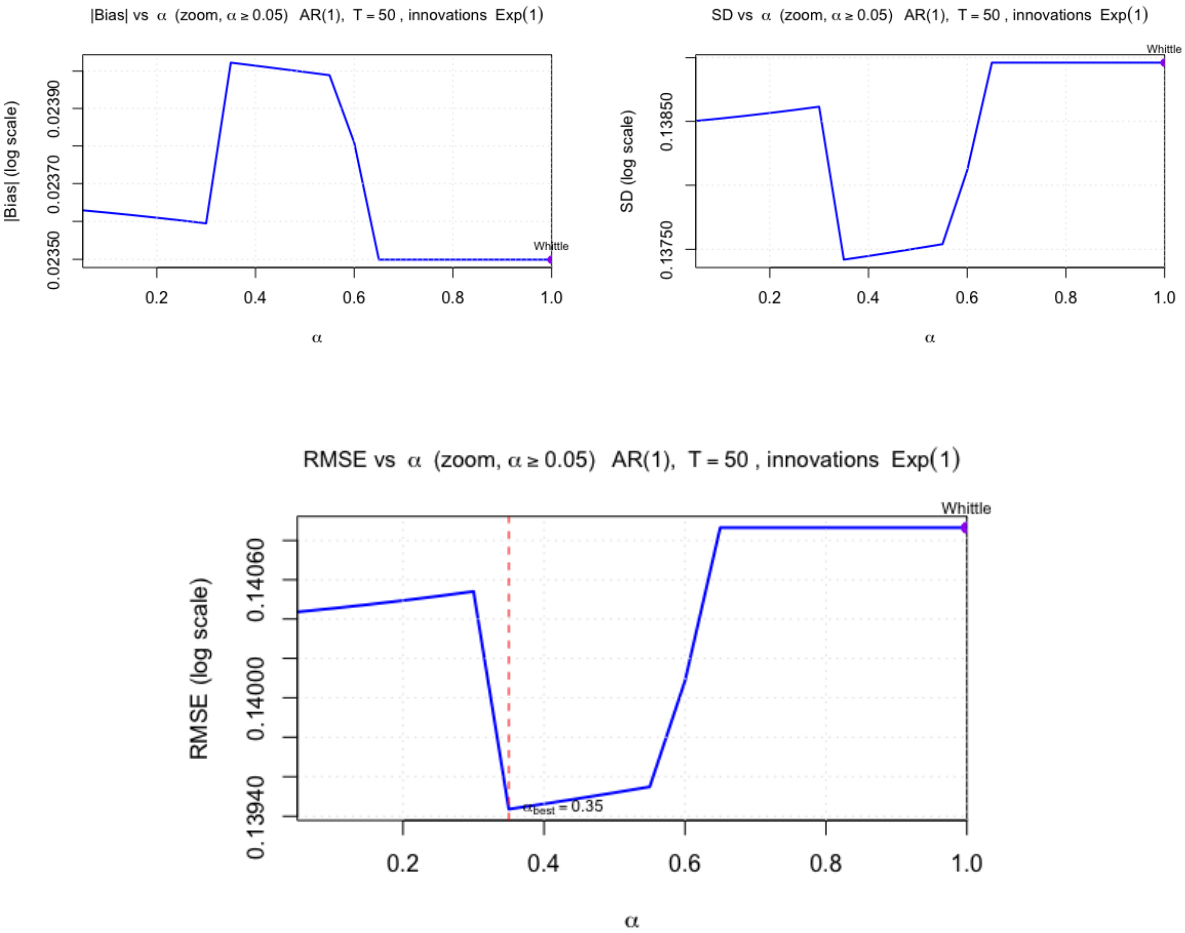


Figure 4.3: Zoomed absolute bias, standard deviation, and RMSE on a logarithmic scale for $\alpha \geq 0.05$, AR(1), $T = 50$, Exp(1) innovations.

Figure 4.3 reveals micro-variations on the order of 10^{-4} to 10^{-3}, confirming that the Hybrid estimator is effectively indistinguishable from the Whittle estimator once α exceeds 0.05.

Figure 4.4 compares the finite-sample distributions of the three estimators. The FDEL estimator exhibits substantial dispersion and frequent outliers, reflecting the strong skewness and heavy tails of the innovations. In contrast, both the Whittle and Hybrid estimators produce tight, approximately symmetric distributions concentrated near the true value $\phi_0 = 0.4$.

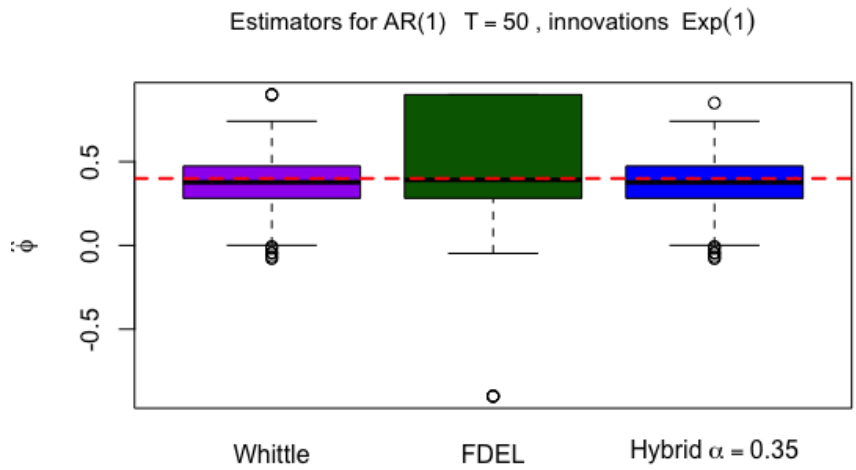


Figure 4.4: Figure 4.4: Finite-sample distributions of the Whittle, FDEL, and Hybrid estimators ($\alpha^* = 0.35$) for the AR(1) model with $T = 50$ and Exp(1) innovations. Boxplots summarize the Monte Carlo distribution of $\widehat{\phi}$ across replications, with the dashed line indicating the true parameter value. The FDEL estimator exhibits substantial dispersion and outliers, while the Hybrid estimator closely matches the Whittle estimator in both location and spread.

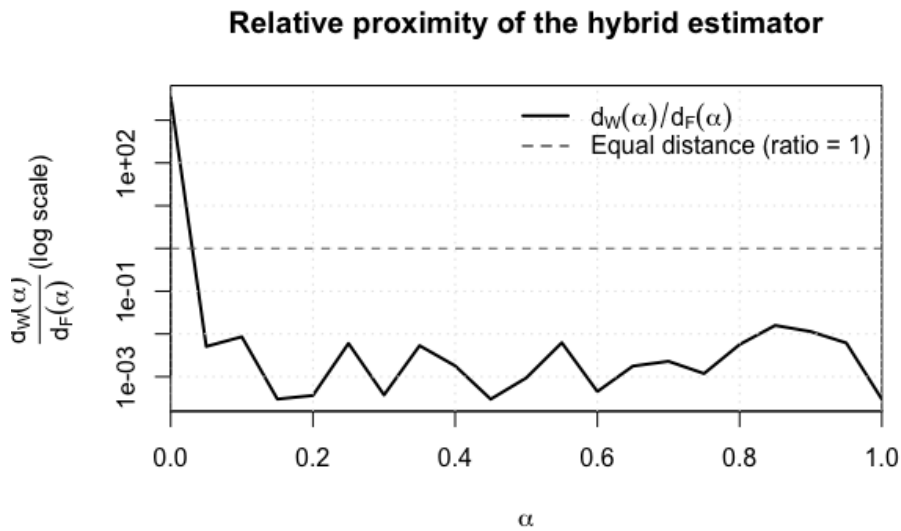


Figure 4.5: Relative proximity of the hybrid estimator to the Whittle and FDEL estimators as a function of the mixing parameter α. The solid curve shows the ratio $R(\alpha) = |\text{median}(\widehat{\phi}_\alpha) - \text{median}(\widehat{\phi}_W)|/|\text{median}(\widehat{\phi}_\alpha) - \text{median}(\widehat{\phi}_F)|$ on a logarithmic scale. The dashed horizontal line at $R(\alpha) = 1$ corresponds to equal distance from the two estimators. Values below one indicate that the hybrid estimator is closer to Whittle than to FDEL.

Dominance diagnostics for the hybrid estimator

To assess how the hybrid estimator is positioned relative to its two components, we consider the relative distance ratio

$$R(\alpha) = \frac{\left|\text{median}(\hat{\phi}_\alpha) - \text{median}(\hat{\phi}_W)\right|}{\left|\text{median}(\hat{\phi}_\alpha) - \text{median}(\hat{\phi}_F)\right|},$$

displayed on a logarithmic scale in Figure 4.5. Values below one indicate that the hybrid estimator is closer to Whittle than to FDEL. Across all simulation settings, $R(\alpha)$ drops sharply below unity for arbitrarily small positive values of α and remains several orders of magnitude below one thereafter.

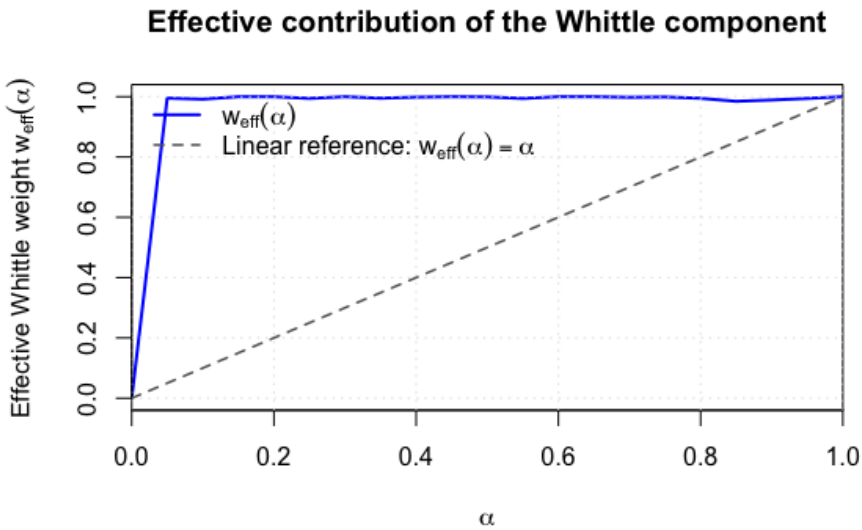


Figure 4.6: Effective contribution of the Whittle component in the hybrid estimator as a function of the mixing parameter α. The solid curve shows the effective Whittle weight $w_{\text{eff}}(\alpha) = \{\text{median}(\hat{\phi}_\alpha) - \text{median}(\hat{\phi}_F)\}/\{\text{median}(\hat{\phi}_W) - \text{median}(\hat{\phi}_F)\}$, computed from the relative location of the hybrid estimator between the FDEL and Whittle estimators. The dashed line represents the linear reference $w_{\text{eff}}(\alpha) = \alpha$. Deviations from the diagonal indicate that the empirical contribution of the Whittle component does not scale linearly with the nominal mixing weight.

Figure 4.6 reports the effective Whittle weight

$$w_{\text{eff}}(\alpha) = \frac{\text{median}(\hat{\phi}_\alpha) - \text{median}(\hat{\phi}_F)}{\text{median}(\hat{\phi}_W) - \text{median}(\hat{\phi}_F)}.$$

Although the hybrid estimator is formally defined as a convex combination, $w_{\text{eff}}(\alpha)$ rapidly approaches one for any $\alpha > 0$. Taken together, these diagnostics indicate that Whittle dominance is a structural feature of the hybrid criterion rather than a numerical artefact of a particular simulation setting.

4.2 ARMA(1,1)

We now turn to the autoregressive moving-average model of order $(1, 1)$, which introduces an additional moving-average component and therefore allows us to assess the behaviour of the estimators in a multivariate parameter setting. The data-generating process is given by

$$X_t = \phi_0 X_{t-1} + \varepsilon_t + \theta_0 \varepsilon_{t-1}, \qquad t = 1, \ldots, T, \tag{4.3}$$

where $(\phi_0, \theta_0) = (0.2, 0.4)$ and $\{\varepsilon_t\}$ denotes an i.i.d. innovation sequence with zero mean and finite variance.

Table 4.3: ARMA(1,1), $(\phi_0, \theta_0) = (0.2, 0.4)$, t_5 innovations

T	Param.	Stat.	Whittle	FDEL	Hybrid
50	ϕ	Bias	0.00915	0.00131	0.00817
		SD	0.26454	0.27104	0.26088
		RMSE	0.26456	0.27091	**0.26087**
	θ	Bias	-0.01049	-0.01341	-0.00926
		SD	0.27472	0.28435	0.27040
		RMSE	0.27478	0.28452	**0.27042**
	$\phi+\theta$	$\text{RMSE}_{\phi+\theta}$	0.53935	0.55543	**0.53130**
					$\alpha^* = 0.40$
75	ϕ	Bias	-0.00294	-0.00289	-0.00289
		SD	0.22318	0.22307	0.22308
		RMSE	0.22308	0.22298	**0.22298**
	θ	Bias	-0.00648	-0.00648	-0.00648
		SD	0.22079	0.22079	0.22079
		RMSE	0.22077	0.22077	0.22077
	$\phi+\theta$	$\text{RMSE}_{\phi+\theta}$	0.44386	**0.44375**	0.44376
					$\alpha^* = 0.05$

Table 4.4: ARMA(1,1), $(\phi_0, \theta_0) = (0.2, 0.4)$, $\chi^2(5)$ innovations

T	Param.	Stat.	Whittle	FDEL	Hybrid
50	ϕ	Bias	0.00439	-0.00466	0.00389
		SD	0.27239	0.27928	0.26921
		RMSE	0.27229	0.27918	**0.26910**
	θ	Bias	-0.01079	-0.01459	-0.01001
		SD	0.28059	0.29134	0.27692
		RMSE	0.28066	0.29156	**0.27696**
	$\phi+\theta$	$\text{RMSE}_{\phi+\theta}$	0.55295	0.57073	**0.54606**
					$\alpha^* = 0.45$
75	ϕ	Bias	-0.01084	-0.01046	-0.01070
		SD	0.22019	0.21905	0.21968
		RMSE	0.22035	**0.21919**	0.21983
	θ	Bias	0.00498	0.00308	0.00489
		SD	0.21476	0.21829	0.21452
		RMSE	0.21471	0.21821	**0.21447**
	$\phi+\theta$	$\text{RMSE}_{\phi+\theta}$	0.43506	0.43739	**0.43430**
					$\alpha^* = 0.05$

Table 4.5: ARMA(1,1), $(\phi_0, \theta_0) = (0.2, 0.4)$, Exp(1) innovations

T	Param.	Stat.	Whittle	FDEL	Hybrid
50	ϕ	Bias	-0.00250	-0.01535	-0.00474
		SD	0.27060	0.28332	0.26810
		RMSE	0.27047	0.28360	**0.26801**
	θ	Bias	-0.00492	-0.01176	-0.00332
		SD	0.27582	0.29571	0.27191
		RMSE	0.27573	0.29580	**0.27180**
	$\phi+\theta$	$\text{RMSE}_{\phi+\theta}$	0.54620	0.57939	**0.53981**
					$\alpha^* = 0.40$
75	ϕ	Bias	-0.01762	-0.01759	-0.01828
		SD	0.22701	0.22619	0.22492
		RMSE	0.22758	0.22676	**0.22554**
	θ	Bias	0.00575	0.00764	0.00680
		SD	0.22977	0.22668	0.22625
		RMSE	0.22973	0.22670	**0.22624**
	$\phi+\theta$	$\text{RMSE}_{\phi+\theta}$	0.45731	0.45346	**0.45179**
					$\alpha^* = 0.05$

Discussion

Tables 4.3 – 4.5 report bias, standard deviation, and RMSE for the individual parameters ϕ and θ, together with the joint performance measure $\text{RMSE}_{\phi+\theta}$. The remaining results are reported in Appendix 9, on Tables 9.3 – 9.6. As in the AR(1) case, the most pronounced differences between estimators occur at the smallest sample size $T = 50$ and under heavy-tailed or skewed innovations. Across the Student-t_5, $\chi^2(5)$, and Exp(1) distributions, the FDEL estimator again exhibits inflated variability, leading to substantially larger joint RMSE values. In contrast, the Hybrid estimator consistently improves upon FDEL and, for all three innovation settings at $T = 50$, attains the smallest $\text{RMSE}_{\phi+\theta}$. The gains are particularly visible under the skewed $\chi^2(5)$ and Exp(1) innovations.

At $T = 75$, differences between the three estimators narrow considerably. All methods display similar bias and variability, and their joint RMSE values become nearly indistinguishable across innovation settings, indicating that the finite-sample advantages of the Hybrid estimator are primarily relevant in the most challenging small-sample regimes.

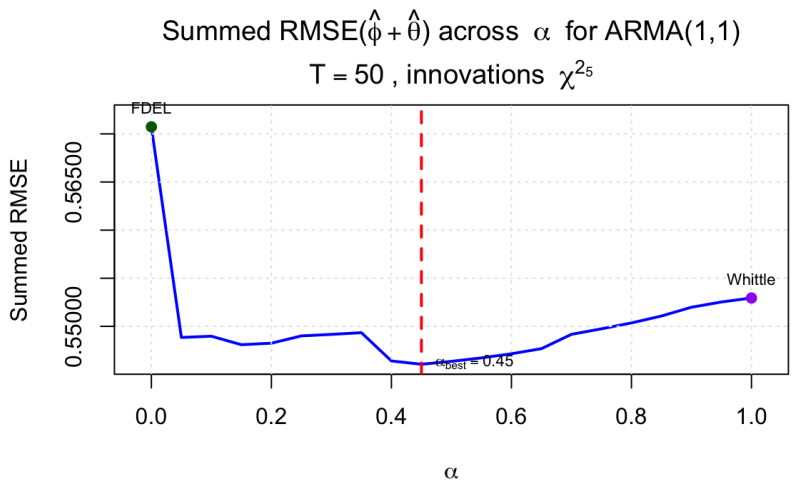
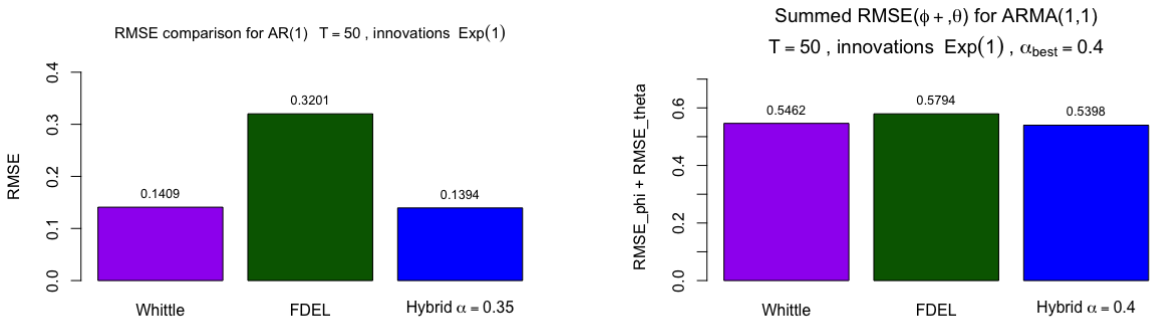


Figure 4.7: Summed RMSE as a function of the mixing parameter α for the ARMA(1,1) model with $T = 50$ and Exp(1) innovations. The curve is representative of the ARMA(1,1) results: the RMSE drops sharply between $\alpha = 0$ and small positive values of α, and then remains essentially flat for $\alpha \geq 0.05$. This behaviour explains the irregular selection of α^* across Monte Carlo replications, while making the location of the minimum visually identifiable without the need for further zooming.

Relative to the AR(1) benchmark, discrepancies between the FDEL estimator and the Whittle and Hybrid estimators are markedly reduced in the ARMA(1,1) setting. Although FDEL remains slightly inferior at $T = 50$, its joint RMSE is substantially closer to that of the competing methods, indicating that the inclusion of a moving-average component mitigates part of the small-sample instability observed in the pure AR(1) case. Moreover, as shown in Tables 9.3 and 9.4, FDEL attains marginally smaller RMSE than both Whittle and Hybrid estimators in several configurations at moderate sample sizes (e.g. $T = 75$ and $T = 100$), suggesting that its relative efficiency can match that of parametric approaches when model dynamics are richer. The difference stays very small.

This comparison is illustrated in Figure 4.8, which contrasts estimator performance under Exp(1) innovations for the AR(1) and ARMA(1,1) models. While the qualitative ranking of the estimators remains unchanged, the performance gap between FDEL and the Whittle and Hybrid estimators is noticeably smaller in the ARMA setting.



(a) AR(1): RMSE_{ϕ} (b) ARMA(1,1): $\text{RMSE}_{\phi+\theta}$

Figure 4.8: Comparison of Whittle, FDEL, and Hybrid estimators at $T = 50$ under Exp(1) innovations for the AR(1) and ARMA(1,1) models.

4.3 ARFIMA$(1, d, 0)$

We turn to the fractionally integrated autoregressive model, which allows for long-range dependence and therefore extends the previous short-memory AR and ARMA settings. The data-generating process is given by

$$(1 - \phi_0 L)(1 - L)^d X_t = \varepsilon_t, \qquad t = 1, \ldots, T, \tag{4.4}$$

where L denotes the lag operator, $\phi_0 = 0.4$, and $\{\varepsilon_t\}$ is an i.i.d. innovation sequence with zero mean and finite variance. The fractional differencing parameter $d \in (0, 1)$ governs the strength of persistence: larger values of d correspond to slower decay of autocorrelations and stronger long-range dependence.

We consider two representative values of the long-memory parameter, namely $d = 0.1$ and $d = 0.4$. The case $d = 0.1$ corresponds to weak long memory and is close to the short-memory boundary, while $d = 0.4$ represents a pronounced long-memory regime. This design allows us to assess how the stabilising effect of the Hybrid estimator interacts with increasing persistence and whether the behaviour observed in the AR and ARMA models persists in the presence of long-range dependence.

Table 4.6: ARFIMA$(1, d, 0)$, $d = 0.1$, t_5 innovations

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.00844	0.00897	-0.00951
	SD	0.14087	0.24607	0.13715
	RMSE	0.14105	0.24611	**0.13741**
				$\alpha^* = 0.80$
75	Bias	-0.00933	-0.00224	-0.00933
	SD	0.11066	0.13556	0.11066
	RMSE	0.11099	0.13551	0.11099
				$\alpha^* = 0.70$
100	Bias	-0.01075	-0.00900	-0.01075
	SD	0.09523	0.10300	0.09523
	RMSE	0.09579	0.10334	0.09579
				$\alpha^* = 0.85$
200	Bias	-0.00576	-0.00531	-0.00576
	SD	0.06887	0.07162	0.06887
	RMSE	0.06908	0.07178	0.06908
				$\alpha^* = 0.80$

Table 4.7: ARFIMA$(1, d, 0)$, $d = 0.1$, $\chi^2(5)$ innovations

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.01417	0.01015	-0.01450
	SD	0.14198	0.23756	0.14082
	RMSE	0.14261	0.23765	**0.14150**
				$\alpha^* = 0.65$
75	Bias	-0.01103	-0.00343	-0.01103
	SD	0.10804	0.14347	0.10804
	RMSE	0.10855	0.14344	0.10855
				$\alpha^* = 0.85$
100	Bias	-0.00872	-0.00676	-0.00872
	SD	0.09786	0.10671	0.09786
	RMSE	0.09820	0.10687	0.09820
				$\alpha^* = 0.70$
200	Bias	-0.00911	-0.00911	-0.00911
	SD	0.06507	0.06507	0.06507
	RMSE	0.06568	0.06568	0.06568
				$\alpha^* = 0.35$

Table 4.8: ARFIMA$(1, d, 0)$, $d = 0.1$, Exp(1) innovations

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.01039	0.01641	-0.01141
	SD	0.13924	0.23359	0.13561
	RMSE	0.13956	0.23405	**0.13602**
				$\alpha^* = 0.70$
75	Bias	-0.01259	0.00110	-0.01259
	SD	0.10879	0.15316	0.10879
	RMSE	0.10946	0.15308	0.10946
				$\alpha^* = 0.75$
100	Bias	-0.00667	-0.00233	-0.00667
	SD	0.09338	0.11255	0.09338
	RMSE	0.09357	0.11252	0.09357
				$\alpha^* = 0.40$
200	Bias	-0.00648	-0.00648	-0.00648
	SD	0.06396	0.06396	0.06396
	RMSE	0.06426	0.06426	0.06426
				$\alpha^* = 0.80$

Discussion: $d = 0.1$

Tables 4.6 – 4.8 report bias, standard deviation, and RMSE for the ARFIMA$(1, d, 0)$ model with mild long memory ($d = 0.1$) under non-Gaussian innovations. The remaining results are reported in Appendix 9, on Table 9.7. The overall picture is consistent with the AR and ARMA results: the clearest separation between methods occurs at the smallest sample size $T = 50$, where the frequency-domain empirical likelihood component is the most fragile.

Across Student-t_5, $\chi^2(5)$, and Exp(1) innovations, the FDEL estimator shows substantially larger dispersion at $T = 50$, which translates directly into noticeably larger RMSE. In the same settings, Whittle remains stable, and the Hybrid estimator tracks

Whittle very closely. For $T = 50$, the Hybrid RMSE is the smallest in all three tables, but the improvement over Whittle is modest in magnitude.

As T increases to 75 and 100, differences narrow quickly. From $T \geq 75$ onward, Whittle and Hybrid are essentially indistinguishable in these experiments, while FDEL improves markedly compared to $T = 50$. By $T = 200$, all three procedures are very close.

Table 4.9: ARFIMA$(1, d, 0)$, $d = 0.4$, t_5 innovations

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.00535	0.00445	-0.00563
	SD	0.13751	0.21230	0.13644
	RMSE	0.13755	0.21224	**0.13649**
				$\alpha^* = 0.25$
75	Bias	-0.00586	0.00171	-0.00586
	SD	0.11042	0.13650	0.11042
	RMSE	0.11052	0.13644	0.11052
				$\alpha^* = 0.70$
100	Bias	-0.00695	-0.00526	-0.00695
	SD	0.09938	0.11598	0.09938
	RMSE	0.09958	0.11605	0.09958
				$\alpha^* = 0.60$
200	Bias	-0.00153	-0.00108	-0.00153
	SD	0.06961	0.07231	0.06961
	RMSE	0.06959	0.07228	0.06959
				$\alpha^* = 0.75$

Table 4.10: ARFIMA$(1, d, 0)$,$d = 0.4$,$\chi^2(5)$ innovations

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.00044	0.01901	-0.00073
	SD	0.13808	0.22563	0.13702
	RMSE	0.13802	0.22632	**0.13696**
				$\alpha^* = 0.65$
75	Bias	0.00248	0.01049	0.00248
	SD	0.11705	0.14346	0.11705
	RMSE	0.11701	0.14377	0.11701
				$\alpha^* = 0.75$
100	Bias	0.00170	0.00476	0.00170
	SD	0.09378	0.10678	0.09378
	RMSE	0.09374	0.10683	0.09374
				$\alpha^* = 0.80$
200	Bias	0.00049	0.00049	0.00049
	SD	0.06933	0.06933	0.06933
	RMSE	0.06929	0.06929	0.06929
				$\alpha^* = 0.80$

Table 4.11: ARFIMA$(1, d, 0)$, $d = 0.4$, Exp(1) innovations

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.00830	0.01933	-0.00981
	SD	0.14790	0.26192	0.14263
	RMSE	0.14805	0.26250	**0.14289**
				$\alpha^* = 0.65$
75	Bias	-0.00243	0.00982	-0.00243
	SD	0.11044	0.15038	0.11044
	RMSE	0.11041	0.15063	0.11041
				$\alpha^* = 0.85$
100	Bias	-0.00159	0.00283	-0.00159
	SD	0.09637	0.11426	0.09637
	RMSE	0.09633	0.11424	0.09633
				$\alpha^* = 0.80$
200	Bias	0.00496	0.00496	0.00496
	SD	0.06407	0.06407	0.06407
	RMSE	0.06423	0.06423	0.06423
				$\alpha^* = 0.90$

Discussion: $d = 0.4$

Tables 4.9 – 4.11 summarise the same comparison for the stronger dependence case $d = 0.4$. The remaining results are reported on Table 9.8. The qualitative ranking is unchanged: at $T = 50$ under non-Gaussian innovations, FDEL continues to have the largest RMSE, while Whittle and Hybrid remain much more stable. Here again, the Hybrid estimator matches Whittle extremely closely, with only small numerical differences (in favour of Hybrid) at $T = 50$.

For $T \geq 75$, the three estimators become progressively harder to distinguish, and by $T = 200$ their reported error measures are nearly identical. As in the previous models, the selected α^* does not display a clear pattern across settings, which is consistent with

the fact that the Hybrid and Whittle performance curves become nearly flat once α is away from zero in these Monte Carlo designs.

4.4 AR(1) with ARCH(1) Volatility

We consider an autoregressive model with conditional heteroskedasticity, which allows us to assess the robustness of the proposed estimators in the presence of time-varying volatility. The data-generating process is given by

$$X_t = \phi_0 X_{t-1} + \varepsilon_t, \qquad t = 1, \ldots, T, \tag{4.5}$$

$$\varepsilon_t = \sigma_t z_t, \tag{4.6}$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2, \tag{4.7}$$

where $\phi_0 = 0.4$, $(\alpha_0, \alpha_1) = (0.5, 0.4)$, and $\{z_t\}$ is an i.i.d. innovation sequence with zero mean and unit variance. This specification corresponds to a standard ARCH(1) model for the conditional variance and introduces conditional heteroskedasticity while preserving linear dependence in the conditional mean.

Unlike the AR, ARMA, and ARFIMA settings considered previously, the ARCH(1) volatility structure introduces a form of model misspecification for the Whittle, FDEL, and Hybrid estimators, which are derived under the assumption of second-order stationarity with constant unconditional variance.

Table 4.12: AR(1)–ARCH(1), t_5 innovations, $T = 50$

Statistic	Whittle	FDEL	Hybrid
Bias	-0.04029	-0.02346	-0.04236
SD	0.18805	0.29984	0.18340
RMSE	0.19222	0.30061	**0.18814**
			$\alpha^* = 0.35$

Table 4.13: AR(1)–ARCH(1), $\chi^2(5)$ innovations, $T = 50$

Statistic	Whittle	FDEL	Hybrid
Bias	-0.05125	-0.04677	-0.05237
SD	0.17850	0.27312	0.17569
RMSE	0.18563	0.27696	**0.18324**
			$\alpha^* = 0.35$

Table 4.14: AR(1)–ARCH(1), Exp(1) innovations, $T = 50$

Statistic	Whittle	FDEL	Hybrid
Bias	-0.06167	-0.05478	-0.06355
SD	0.17375	0.26937	0.16887
RMSE	0.18429	0.27475	**0.18035**
			$\alpha^* = 0.40$

Discussion

Tables 4.12 – 4.14 focus on the most informative configurations, namely those cases in which the Hybrid estimator improves upon both Whittle and FDEL in terms of RMSE. As in previous sections, these differences arise primarily in small samples ($T = 50$) under non-Gaussian innovations; the remaining results are reported in Appendix 9, on Tables 9.9 – 9.12.

Across innovation distributions, the FDEL estimator again exhibits inflated variability in small samples. This effect is more pronounced in the presence of conditional heteroskedasticity: under ARCH-type volatility, FDEL can display substantially larger RMSE values than its competitors, suggesting that the additional departure from the "ideal" frequency-domain approximation is costly in finite samples. In contrast, the Whittle estimator remains comparatively stable despite the volatility misspecification. The Hybrid estimator shows the same stabilising behaviour seen in the AR, ARMA, and ARFIMA experiments. In practice, assigning even a small positive weight to the Whittle contrast is enough to dampen the variance inflation that characterises FDEL at $\alpha = 0$. In the reported $T = 50$ cases under heavy-tailed or skewed innovations, the Hybrid estimator attains the smallest RMSE, with improvements over Whittle that are modest but persistent in the most adverse settings.

Figure 4.9 shows RMSE as a function of α for the AR(1)–ARCH(1) model under Exp(1) innovations at $T = 50$. The improvement when moving away from $\alpha = 0$ is particularly visible here: the Hybrid estimator achieves a smaller RMSE than pure FDEL

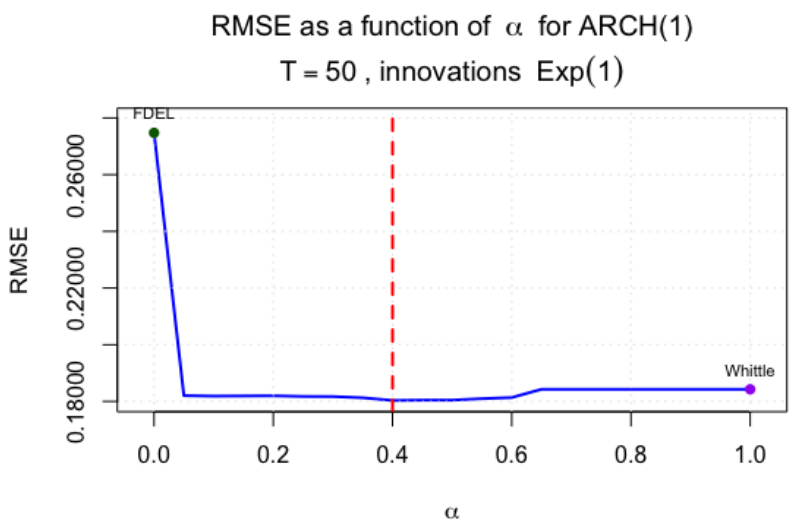


Figure 4.9: RMSE as a function of the mixing parameter α for the AR(1)–ARCH(1) model with Exp(1) innovations at $T = 50$. The sharp drop between $\alpha = 0$ and small positive values illustrates the reduction in FDEL-driven variance inflation once the Whittle component is introduced; the minimum occurs in a region where the curve is already relatively flat.

and can remain slightly below Whittle across a non-trivial range of α values. While the absolute differences are not large, they are clearly visible on the original scale, which is not always the case in the linear homoskedastic experiments.

As the sample size increases, differences between the three estimators diminish rapidly. For $T \geq 100$, in Appendix on Tables 9.9 – 9.12, all methods become very close in terms of bias, SD, and RMSE, and the empirically selected tuning parameters α^* again show no consistent pattern. This mirrors earlier sections and is consistent with the RMSE surface being almost flat once the Whittle component receives a non-zero weight. Overall, the ARCH(1) experiment reinforces the same message as the preceding models: FDEL is the method that is most sensitive to the difficult finite-sample regimes, and the Hybrid criterion mainly acts as a stabiliser in those cases, while becoming nearly indistinguishable from Whittle as T grows.

4.5 AR(1) with Additive Outliers

We investigate the robustness of the competing estimators in the presence of additive outliers, which represent a classical form of model contamination and are known to severely affect estimation procedures based on second-order moments. The baseline data-generating process is the AR(1) model

$$X_t = \phi_0 X_{t-1} + \varepsilon_t, \qquad t = 1, \ldots, T, \tag{4.8}$$

with $\phi_0 = 0.4$ and $\{\varepsilon_t\}$ an i.i.d. innovation sequence with zero mean and finite variance. We contaminate the observed process by replacing a fraction of the observations with fixed additive shocks. Specifically, the observed series is given by

$$X_t^{\text{out}} = X_t + \sum_{i=1}^{k} c_i \mathbf{1}_{\{t=x_i\}}, \tag{4.9}$$

where $\mathbf{1}_{\{t=x_i\}}$ denotes the indicator function selecting the contaminated time points $\{x_1, \ldots, x_k\}$, and $\{c_i\}$ are fixed contamination magnitudes. In our simulation design, the contamination proportion is set to $k/T = 5\%$, so that five percent of the observations are affected by additive outliers. The locations of the outliers are drawn uniformly at random, independently of the underlying process.

Discussion

Table 4.15 reports the bias, standard deviation, and RMSE of the Whittle, FDEL, and Hybrid estimators for the AR(1) model under 5% additive outlier contamination. Results for alternative innovation distributions are reported in Appendix 9, see Tables 9.13 and 9.14. For brevity and layout considerations, results under Gaussian innovations are omitted from the appendix, as they exhibit patterns analogous to those observed for the other distributions. In contrast to the heavy-tailed, heteroskedastic, and long-memory settings examined earlier, additive contamination primarily induces a large systematic bias that dominates the overall estimation error across all sample sizes. The bias compo-

Table 4.15: Bias, standard deviation and RMSE of Whittle, FDEL and Hybrid estimators for AR(1) with $\phi_0 = 0.4$ contaminated by 5% additive outliers under Exp(1) innovations.

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.16854	-0.14714	-0.16958
	SD	0.16197	0.21190	0.15879
	RMSE	0.23370	0.25789	**0.23226**
				$\alpha^* = 0.60$
75	Bias	-0.21541	-0.21296	-0.21541
	SD	0.12850	0.13776	0.12850
	RMSE	0.25079	0.25360	0.25079
				$\alpha^* = 0.95$
100	Bias	-0.20508	-0.20474	-0.20508
	SD	0.11604	0.11757	0.11604
	RMSE	0.23560	0.23607	0.23560
				$\alpha^* = 0.05$
200	Bias	-0.20669	-0.20669	-0.20669
	SD	0.08156	0.08156	0.08156
	RMSE	0.22218	0.22218	0.22218
				$\alpha^* = 0.05$

nent is substantial relative to the reported Monte Carlo standard deviations, so increases in T do not yield the same improvements in RMSE observed under correctly specified models.

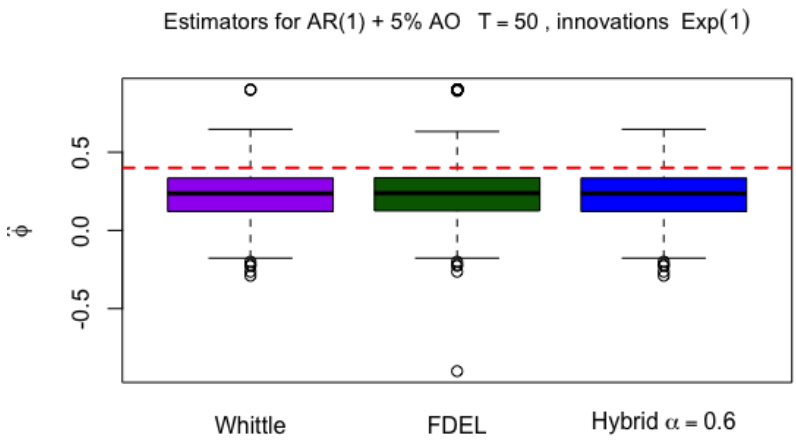


Figure 4.10: Finite-sample distributions of the Whittle, FDEL, and Hybrid estimators ($\alpha = 0.6$) for the AR(1) model contaminated by 5% additive outliers, with $T = 50$ and Exp(1) innovations. The dashed horizontal line denotes the true parameter value $\phi_0 = 0.4$. All three estimators exhibit substantial downward bias, while differences in dispersion are comparatively minor, illustrating that additive contamination primarily induces systematic bias rather than variance inflation.

Figure 4.10 provides a visual summary of this effect for the representative case $T = 50$

under Exp(1) innovations. The boxplots show that the sampling distributions of all three estimators are markedly shifted away from the true parameter value $\phi_0 = 0.4$. Unlike earlier experiments, here the dispersion of the estimators is comparatively similar, while the entire distributions are displaced downward. This confirms that the impact of additive outliers operates primarily through bias rather than through instability or excessive variability.

The Whittle and Hybrid estimators are effectively indistinguishable, indicating once again that for $\alpha > 0$ the Hybrid criterion collapses onto the Whittle solution. The FDEL estimator exhibits a slightly smaller bias in some configurations, but this is offset by higher dispersion, leading to larger RMSE values overall.

Overall, the additive-outlier experiment clarifies an important limitation shared by the Whittle and Hybrid approaches. While the Hybrid estimator is effective at regularising FDEL-driven small-sample variability under non-Gaussian innovations, volatility misspecification, and long-range dependence, it does not provide robustness against gross additive contamination. Addressing such bias would require estimation strategies explicitly designed for outlier resistance, which lies beyond the scope of the present framework.

4.6 AR(1) with Skewed Heavy-Tailed Innovations

We now examine the finite-sample behaviour of the competing estimators when the innovation distribution is both heavy-tailed and asymmetric. The data-generating process is again the AR(1) model

$$X_t = \phi_0 X_{t-1} + \varepsilon_t, \qquad t = 1, \ldots, T, \tag{4.10}$$

with $\phi_0 = 0.4$. The innovation sequence $\{\varepsilon_t\}$ follows a split-t distribution with $\nu = 5$ degrees of freedom and asymmetry parameters $(a, b) = (1.6, 0.8)$, generating right-skewed, heavy-tailed innovations while preserving finite second moments.

Table 4.16: Bias, standard deviation and RMSE of Whittle, FDEL and Hybrid estimator for AR(1) with $\phi_0 = 0.4$ under skewed heavy-tailed innovations (split-t with $\nu = 5$, $a = 1.6$, $b = 0.8$).

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.02702	0.08526	-0.02702
	SD	0.12841	0.27756	0.12841
	RMSE	0.13116	0.29022	0.13116
				$\alpha^* = 0.50$
75	Bias	-0.01578	0.06293	-0.01578
	SD	0.10595	0.23372	0.10595
	RMSE	0.10706	0.24193	0.10706
				$\alpha^* = 0.40$
100	Bias	-0.00759	0.06040	-0.00759
	SD	0.09317	0.21378	0.09317
	RMSE	0.09343	0.22204	0.09343
				$\alpha^* = 0.60$
200	Bias	-0.00707	0.01454	-0.00707
	SD	0.06453	0.13013	0.06453
	RMSE	0.06489	0.13088	0.06489
				$\alpha^* = 0.45$
300	Bias	-0.00565	0.00222	-0.00565
	SD	0.05327	0.08736	0.05327
	RMSE	0.05355	0.08735	0.05355
				$\alpha^* = 0.10$

Discussion

Table 4.16 reports bias, standard deviation, and RMSE for the Whittle, FDEL, and Hybrid estimators under skewed heavy-tailed innovations. The qualitative picture closely mirrors that obtained under symmetric heavy-tailed distributions, with the most pronounced differences occurring in small samples.

At $T = 50$ and $T = 75$, the FDEL estimator exhibits substantial bias and pronounced variance inflation, resulting in RMSE values more than twice those of the Whittle estimator. This instability persists despite the innovations having finite second moments, indicating that the combination of heavy tails and asymmetry is particularly detrimental for the empirical likelihood component in small samples. In contrast, the Whittle estimator remains numerically stable across all sample sizes.

The Hybrid estimator coincides exactly with the Whittle estimator in all reported

configurations, yielding identical bias, standard deviation, and RMSE. As in earlier sections, this reflects the immediate dominance of the Whittle component once a positive mixing weight is introduced. Consequently, the empirically selected tuning parameters α^* display no systematic pattern and reflect numerical indifference rather than a meaningful bias–variance trade-off.

4.7 Model Misspecification: AR(2) DGM, AR(1) fit

To assess the behaviour of the proposed estimators under model misspecification, we consider a setting in which the fitted model does not coincide with the true data generating mechanism. Specifically, the observed process $\{X_t\}$ is generated from a stationary AR(2) model,

$$X_t = \phi_{1,0} X_{t-1} + \phi_{2,0} X_{t-2} + \varepsilon_t, \qquad (\phi_{1,0}, \phi_{2,0}) = (0.6, -0.3), \tag{4.11}$$

where $\{\varepsilon_t\}$ denotes a sequence of independent innovations with zero mean and unit variance, drawn from Gaussian, Student-t_5, $\chi^2(5)$ (standardised), or Exp(1) (centred) distributions. Despite the true AR(2) structure, we deliberately fit an AR(1) model,

$$X_t = \phi X_{t-1} + u_t, \tag{4.12}$$

using the Whittle, FDEL, and Hybrid estimators. This framework corresponds to fitting a misspecified model

$$X_t = G'(\ldots, Z_{t-1}, Z_t, \boldsymbol{\theta})$$

when the true process is

$$X_t = G(\ldots, Z_{t-1}, Z_t, \boldsymbol{\theta}_0),$$

with $\boldsymbol{\theta}_0 = (\phi_{1,0}, \phi_{2,0})$ and $\boldsymbol{\theta} = \phi$.

Since the usual parameter error metrics are no longer meaningful under misspecification, estimator performance is evaluated in terms of spectral approximation accuracy. In particular, we measure the discrepancy between the true AR(2) spectral density $f_{\boldsymbol{\theta}_0}(\lambda)$

and the fitted AR(1) spectral density $f_{\hat{\boldsymbol{\theta}}}(\lambda)$ via the spectral distance

$$D(\hat{\boldsymbol{\theta}}) = \int_0^{\pi} \left| f_{\boldsymbol{\theta}_0}(\lambda) - f_{\hat{\boldsymbol{\theta}}}(\lambda) \right| d\lambda, \tag{4.13}$$

which is approximated numerically on a fine frequency grid using the trapezoidal rule. Smaller values of $D(\hat{\boldsymbol{\theta}})$ indicate a closer spectral match between the misspecified fitted model and the true data generating process.

This criterion allows for a direct comparison of Whittle, FDEL, and Hybrid estimators in terms of their ability to recover the dominant second-order features of the underlying process, even when the fitted parametric form is incorrect.

Table 4.17: **Representative results.** Mean spectral distance under AR(2) misspecification: t_5 innovations.

T	Whittle	FDEL	Hybrid
50	0.28943	1.19820	**0.28625**
			$\alpha^* = 0.55$
75	0.27903	1.08848	0.27903
			$\alpha^* = 0.45$
100	0.27079	1.02617	0.27079
			$\alpha^* = 0.30$
200	0.26598	0.88099	0.26598
			$\alpha^* = 0.25$

Table 4.18: **Representative results.** Mean spectral distance under AR(2) misspecification: Exp(1) innovations.

T	Whittle	FDEL	Hybrid
50	0.29085	1.21914	**0.28544**
			$\alpha^* = 0.50$
75	0.27762	1.13331	0.27762
			$\alpha^* = 0.10$
100	0.27142	1.03049	0.27142
			$\alpha^* = 0.20$
200	0.26566	0.86878	0.26566
			$\alpha^* = 0.35$

Discussion

Table 4.17 reports the mean spectral distance between the true AR(2) spectrum and the fitted AR(1) spectrum for the Whittle, FDEL, and Hybrid estimators under model misspecification. Several systematic patterns are observed across innovation distributions and sample sizes.

Across all reported configurations, the Whittle estimator yields relatively small spectral distances that decrease gradually as T increases. In contrast, the FDEL estimator

produces substantially larger spectral distances, particularly in small samples and under non-Gaussian innovations. The magnitude of these differences indicates that the empirical likelihood-based approach is more sensitive to departures from the assumed parametric structure in this misspecified setting.

The Hybrid estimator closely tracks the Whittle estimator throughout. In some cases, it attains marginally smaller spectral distances, though these differences are numerically small. As in earlier sections, the empirically selected tuning parameter α^* does not exhibit a clear or systematic pattern across sample sizes or innovation distributions.

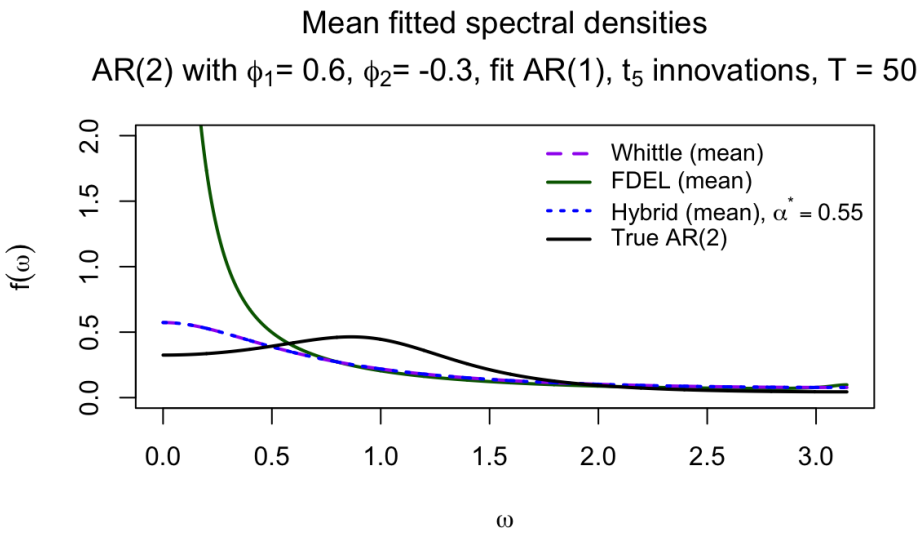


Figure 4.11: Mean fitted spectral densities for an AR(2) data-generating process with $(\phi_1, \phi_2) = (0.6, -0.3)$, fitted using a misspecified AR(1) model, t_5 innovations, and $T = 50$. The Whittle and Hybrid fits are nearly indistinguishable and provide a smooth approximation to the dominant low- and mid-frequency features of the true spectrum despite misspecification. By contrast, the FDEL fit displays a pronounced concentration of power at low frequencies, resulting in a spectral shape that deviates substantially from the true AR(2) spectrum.

Figure 4.11 provides a graphical illustration of these patterns for the representative case $T = 50$ with t_5 innovations. The Whittle and Hybrid fitted spectra are nearly indistinguishable and offer a smooth approximation to the true spectral shape, capturing its main second-order features despite the incorrect AR(1) specification. In contrast, the FDEL fit is characterised by a pronounced inflation of low-frequency power, leading to a spectral shape that departs markedly from the true spectrum. This visual discrepancy is

consistent with the larger spectral distances reported for FDEL in Table 4.17.

As in the correctly specified experiments, the values of α^* selected by minimising the spectral distance tend to lie in regions where the Hybrid estimator exhibits performance very close to that of the Whittle estimator. This suggests that, under misspecification, a wide range of mixing weights yields similar spectral approximations once a non-zero Whittle component is included.

The additional results reported in Appendix 9 (Tables 9.15 and 9.16) indicate that these patterns persist across the remaining innovation distributions. Overall, under AR(2) misspecification, the Hybrid estimator behaves similarly to the Whittle estimator in terms of spectral approximation accuracy, while the FDEL estimator exhibits noticeably poorer performance in this setting.

5 Conclusion

This thesis investigated a Hybrid frequency-domain estimator that combines the Whittle likelihood with frequency-domain empirical likelihood (FDEL), with the aim of assessing whether hybridisation can improve finite-sample performance in settings where empirical likelihood methods are known to be fragile. A comprehensive Monte Carlo study was conducted across a wide range of data-generating mechanisms, including AR, ARMA, and ARFIMA models, conditional heteroskedasticity, additive outliers, skewed heavy-tailed innovations, and model misspecification.

The simulation results show that, across most settings, the Hybrid estimator behaves almost identically to the Whittle estimator once a positive weight is assigned to the Whittle component. This pattern is highly systematic and appears consistently across sample sizes, innovation distributions, and model classes. In particular, even very small values of the tuning parameter α are sufficient to move the Hybrid estimator onto a performance plateau that is numerically indistinguishable from Whittle in terms of RMSE or spectral distance. As a consequence, the empirically selected oracle tuning parameter α^* exhibits irregular and non-informative behaviour across simulation designs, which can

be traced to the near-flatness of the error surfaces for $\alpha > 0$.

Detailed diagnostics in the AR(1) setting provide additional insight into this phenomenon. Although the Hybrid criterion is formally defined as a convex combination of the Whittle and FDEL contrasts, relative-distance and effective-weight diagnostics indicate that the Hybrid estimator becomes much closer to Whittle than to FDEL almost immediately after α departs from zero. These findings suggest that, in finite samples, the Whittle component dominates the hybrid objective and that hybridisation does not lead to a smooth or interpretable interpolation between the two estimation principles.

The FDEL estimator, by contrast, frequently exhibits inflated variability and sensitivity to extreme observations, particularly in small samples and under non-Gaussian innovations, conditional heteroskedasticity or model misspecification. In these settings, FDEL often performs substantially worse than Whittle in terms of RMSE or spectral approximation accuracy. The Hybrid estimator primarily mitigates these extreme behaviours by anchoring the estimation to the Whittle component, rather than by introducing a genuinely distinct compromise between likelihood-based and empirical likelihood-based information.

Instances in which the Hybrid estimator outperforms Whittle are rare and confined to the most challenging configurations, typically involving very small samples combined with heavy-tailed innovations, conditional heteroskedasticity or model misspecification. Even in these cases, the observed improvements are modest in magnitude and diminish rapidly as the sample size increases. For moderate to large T, all three estimators become practically indistinguishable in the reported performance measures.

It is important to emphasise that the tuning parameter α is selected using an oracle Monte Carlo criterion rather than a data-driven procedure. Specifically, α^* is chosen from a fixed grid as the value minimising the Monte Carlo RMSE or spectral distance computed using knowledge of the true data-generating process. This procedure provides an idealised benchmark rather than a feasible selection rule. The fact that Hybrid rarely improves upon Whittle even under such oracle tuning reinforces the conclusion that the observed behaviour reflects structural properties of the hybrid objective, rather than

limitations of the tuning strategy.

Overall, the results of this thesis do not suggest that the Hybrid estimator should be viewed as a general improvement over the Whittle estimator. Instead, they indicate that hybridisation primarily acts as a stabilisation mechanism for empirical likelihood in adverse finite-sample regimes, while offering limited scope for systematic efficiency gains relative to Whittle. From this perspective, the main contribution of the study is to clarify when and how hybridisation alters estimator behaviour, and equally importantly when it does not.

Several directions for future research emerge from these findings. First, the pronounced dominance of the Whittle component observed in finite samples suggests that further progress may require addressing the difference in scale and curvature between the Whittle and FDEL contrasts. Alternative normalisations or curvature-adjusted hybrid criteria could potentially allow the empirical likelihood component to play a more substantive role. Second, the behaviour of the Hybrid estimator in very small samples merits further investigation, as this is the regime in which any gains over Whittle are most likely to arise.

Finally, the selection of the tuning parameter α remains an open problem. In the i.i.d. setting, Dæhlen and Hjort (2024) propose focus-driven hybrid likelihoods based on the Focused Information Criterion (FIC), which target specific functionals rather than global loss measures. Developing an analogous focus-based selection principle in the frequency domain provide a principled framework for hybridisation in time-series applications. While such extensions lie beyond the scope of the present thesis, they offer a natural direction for future work.

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7 Use of AI Tools

During the preparation of this thesis, I used generative artificial intelligence tools, including ChatGPT, as support for R code editing, clarification of technical exposition, and stylistic refinement of draft text. These tools were also used to assist with LaTeX formatting and to help reorganise sections for clarity. All methodological choices, theoretical developments, simulation designs, computational implementations, and interpretations of results are entirely my own. I critically reviewed and validated all AI-assisted content to ensure its accuracy and consistency with the objectives of the thesis.

For transparency, documentation and usage policies for the AI tools consulted are available at:

- https://chatgpt.com/share/694927b5-ca4c-8003-a876-6d793e1d6715
- https://chatgpt.com/share/6949280d-0574-8003-b005-3791595240cc
- https://chatgpt.com/share/69492958-abb4-8003-8ce8-829778abaffa
- https://chatgpt.com/share/694929cd-83e8-8003-9a4b-34f371250dca

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9 Appendix

This appendix reports additional Monte Carlo results that complement those presented in the main text. Its primary purpose is transparency: to document the numerical behaviour of the estimators across all remaining simulation configurations and to support the qualitative comparisons discussed in the core chapters.

In contrast to the main text, which focuses on the most informative non-Gaussian settings, the appendix also includes Gaussian innovations as explicit benchmarks. These benchmark cases provide a reference scale for finite-sample variability under well-behaved data-generating mechanisms and help contextualise the effects observed under heavy tails, skewness, conditional heteroskedasticity, additive outliers and model misspecification.

The tables collected here consist of the simulation results that were not displayed in the main text for space reasons. They are not intended as selective or representative examples, but rather as a complete supplement that allows the reader to verify that the patterns described in the discussion sections extend to the full set of Monte Carlo experiments.

AR(1): Additional Simulation Results

Table 9.1: **Benchmark results.** AR(1): $\mathcal{N}(0,1)$ innovations.

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.02158	0.08748	-0.02158
	SD	0.13852	0.27601	0.13852
	RMSE	0.14012	0.28941	0.14012
				$\alpha^* = 0.55$
75	Bias	-0.01047	0.07597	-0.01047
	SD	0.10400	0.23675	0.10400
	RMSE	0.10448	0.24853	0.10448
				$\alpha^* = 0.65$
100	Bias	-0.00949	0.05623	-0.00949
	SD	0.09759	0.21317	0.09759
	RMSE	0.09800	0.22036	0.09800
				$\alpha^* = 0.10$
200	Bias	-0.00660	0.01825	-0.00660
	SD	0.06617	0.13699	0.06617
	RMSE	0.06646	0.13813	0.06646
				$\alpha^* = 0.25$

Table 9.2: **Representative results.** AR(1): $\chi^2(5)$ innovations.

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.01962	0.11245	-0.01962
	SD	0.13841	0.29022	0.13841
	RMSE	0.13973	0.31110	0.13973
				$\alpha^* = 0.10$
75	Bias	-0.01610	0.07932	-0.01610
	SD	0.10434	0.24509	0.10434
	RMSE	0.10552	0.25749	0.10552
				$\alpha^* = 0.70$
100	Bias	-0.01567	0.06152	-0.01567
	SD	0.09228	0.22281	0.09228
	RMSE	0.09356	0.23104	0.09356
				$\alpha^* = 0.35$
200	Bias	-0.00199	0.04349	-0.00199
	SD	0.06628	0.16999	0.06628
	RMSE	0.06628	0.17538	0.06628
				$\alpha^* = 0.60$

ARMA(1,1): Additional Simulation Results

Table 9.3: **Benchmark results.** ARMA(1,1), $(\phi_0, \theta_0) = (0.2, 0.4)$: $\mathcal{N}(0,1)$ innovations.

T	Parameter	Statistic	Whittle	FDEL	Hybrid
50	ϕ	Bias	-0.01639	-0.01969	-0.01673
	ϕ	SD	0.27628	0.27637	0.27465
	ϕ	RMSE	0.27663	0.27693	**0.27502**
	θ	Bias	0.01209	0.01371	0.01297
	θ	SD	0.28866	0.28716	0.28555
	θ	RMSE	0.28877	0.28734	**0.28571**
	$\phi+\theta$	$\text{RMSE}_{\phi+\theta}$	0.56540	0.56427	**0.56073**
					$\alpha^* = 0.05$
75	ϕ	Bias	-0.01250	-0.01232	-0.01233
	ϕ	SD	0.22442	0.22409	0.22410
	ϕ	RMSE	0.22466	**0.22432**	0.22433
	θ	Bias	0.00912	0.00896	0.00896
	θ	SD	0.22056	0.22023	0.22024
	θ	RMSE	0.22064	**0.22030**	0.22031
	$\phi+\theta$	$\text{RMSE}_{\phi+\theta}$	0.44530	**0.44462**	0.44464
					$\alpha^* = 0.05$
100	ϕ	Bias	-0.00673	-0.00673	-0.00673
	ϕ	SD	0.18774	0.18774	0.18774
	ϕ	RMSE	0.18776	0.18776	0.18776
	θ	Bias	0.00650	0.00650	0.00650
	θ	SD	0.18437	0.18437	0.18437
	θ	RMSE	0.18439	0.18439	0.18439
	$\phi+\theta$	$\text{RMSE}_{\phi+\theta}$	0.37216	0.37216	0.37216
					$\alpha^* = 0.20$
200	ϕ	Bias	-0.00325	-0.00325	-0.00325
	ϕ	SD	0.12492	0.12492	0.12492
	ϕ	RMSE	0.12489	0.12489	0.12489
	θ	Bias	0.00402	0.00402	0.00402
	θ	SD	0.11726	0.11726	0.11726
	θ	RMSE	0.11727	0.11728	0.11727
	$\phi+\theta$	$\text{RMSE}_{\phi+\theta}$	0.24217	0.24217	0.24217
					$\alpha^* = 0.95$

Table 9.4: **Representative results.** ARMA(1,1), $(\phi_0, \theta_0) = (0.2, 0.4)$: t_5 innovations.

T	Parameter	Statistic	Whittle	FDEL	Hybrid
100	ϕ	Bias	-0.01864	-0.01849	-0.01859
	ϕ	SD	0.19302	0.19253	0.19289
	ϕ	RMSE	0.19383	**0.19332**	0.19368
	θ	Bias	0.01393	0.01377	0.01387
	θ	SD	0.19362	0.19322	0.19347
	θ	RMSE	0.19403	**0.19361**	0.19387
	$\phi+\theta$	$\text{RMSE}_{\phi+\theta}$	0.38785	**0.38693**	0.38755
					$\alpha^* = 0.05$
200	ϕ	Bias	-0.00665	-0.00665	-0.00665
	ϕ	SD	0.13162	0.13162	0.13162
	ϕ	RMSE	0.13172	0.13172	0.13172
	θ	Bias	0.00317	0.00317	0.00317
	θ	SD	0.12429	0.12429	0.12429
	θ	RMSE	0.12427	0.12427	0.12427
	$\phi+\theta$	$\text{RMSE}_{\phi+\theta}$	0.25599	0.25599	0.25599
					$\alpha^* = 0.25$

Table 9.5: **Representative results.** ARMA(1,1), $(\phi_0, \theta_0) = (0.2, 0.4)$: $\chi^2(5)$ innovations.

T	Parameter	Statistic	Whittle	FDEL	Hybrid
100	ϕ	Bias	-0.00649	-0.00649	-0.00649
	ϕ	SD	0.18644	0.18644	0.18644
	ϕ	RMSE	0.18646	0.18646	0.18646
	θ	Bias	-0.00137	-0.00137	-0.00137
	θ	SD	0.18521	0.18521	0.18521
	θ	RMSE	0.18512	0.18512	0.18512
	$\phi + \theta$	$\text{RMSE}_{\phi+\theta}$	0.37158	0.37158	0.37158
					$\alpha^* = 0.95$
200	ϕ	Bias	-0.01017	-0.01017	-0.01017
	ϕ	SD	0.12617	0.12617	0.12617
	ϕ	RMSE	0.12652	0.12652	0.12652
	θ	Bias	0.00575	0.00575	0.00575
	θ	SD	0.12197	0.12197	0.12197
	θ	RMSE	0.12204	0.12204	0.12204
	$\phi + \theta$	$\text{RMSE}_{\phi+\theta}$	0.24856	0.24856	0.24856
					$\alpha^* = 0.85$

Table 9.6: **Representative results.** ARMA(1,1), $(\phi_0, \theta_0) = (0.2, 0.4)$: Exp(1) innovations.

T	Parameter	Statistic	Whittle	FDEL	Hybrid
100	ϕ	Bias	-0.01022	-0.01022	-0.01022
	ϕ	SD	0.19496	0.19496	0.19496
	ϕ	RMSE	0.19513	0.19513	0.19513
	θ	Bias	0.00035	0.00035	0.00035
	θ	SD	0.19448	0.19448	0.19448
	θ	RMSE	0.19439	0.19439	0.19439
	$\phi + \theta$	$\text{RMSE}_{\phi+\theta}$	0.38951	0.38952	0.38951
					$\alpha^* = 0.95$
200	ϕ	Bias	0.00037	0.00037	0.00037
	ϕ	SD	0.12522	0.12522	0.12522
	ϕ	RMSE	0.12516	0.12516	0.12516
	θ	Bias	-0.00409	-0.00409	-0.00409
	θ	SD	0.12410	0.12410	0.12410
	θ	RMSE	0.12410	0.12410	0.12410
	$\phi + \theta$	$\text{RMSE}_{\phi+\theta}$	0.24926	0.24926	0.24926
					$\alpha^* = 0.65$

ARFIMA$(1, d, 0)$: Additional Simulation Results

Table 9.7: **Benchmark results.** ARFIMA$(1, d, 0)$ with $d = 0.1$: $\mathcal{N}(0, 1)$ innovations.

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.02027	-0.00501	-0.02027
	SD	0.14114	0.21511	0.14114
	RMSE	0.14252	0.21506	0.14252
				$\alpha^* = 0.65$
75	Bias	-0.00804	-0.00146	-0.00804
	SD	0.11637	0.14646	0.11637
	RMSE	0.11659	0.14640	0.11659
				$\alpha^* = 0.75$
100	Bias	-0.01229	-0.01153	-0.01229
	SD	0.09640	0.10005	0.09640
	RMSE	0.09714	0.10066	0.09714
				$\alpha^* = 0.80$
200	Bias	-0.00726	-0.00726	-0.00726
	SD	0.06786	0.06786	0.06786
	RMSE	0.06821	0.06821	0.06821
				$\alpha^* = 0.65$

Table 9.8: **Benchmark results.** ARFIMA$(1, d, 0)$ with $d = 0.4$: $\mathcal{N}(0, 1)$ innovations.

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.01002	0.00201	-0.01002
	SD	0.13845	0.19676	0.13845
	RMSE	0.13874	0.19668	0.13874
				$\alpha^* = 0.65$
75	Bias	-0.00373	0.00016	-0.00373
	SD	0.11061	0.12553	0.11061
	RMSE	0.11061	0.12547	0.11061
				$\alpha^* = 0.50$
100	Bias	-0.00429	-0.00309	-0.00429
	SD	0.09748	0.10291	0.09748
	RMSE	0.09753	0.10290	0.09753
				$\alpha^* = 0.85$
200	Bias	0.00191	0.00191	0.00191
	SD	0.06815	0.06815	0.06815
	RMSE	0.06814	0.06814	0.06814
				$\alpha^* = 0.10$

AR(1) with ARCH(1) Volatility: Additional Results

Table 9.9: **Benchmark results.** AR(1)–ARCH(1): $\mathcal{N}(0,1)$ innovations. Included as a representative Gaussian benchmark under volatility misspecification.

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.04383	-0.03450	-0.04383
	SD	0.18106	0.26871	0.18106
	RMSE	0.18620	0.27078	0.18620
				$\alpha^* = 0.10$
75	Bias	-0.02847	-0.02271	-0.02847
	SD	0.14647	0.17750	0.14647
	RMSE	0.14914	0.17886	0.14914
				$\alpha^* = 0.80$
100	Bias	-0.02790	-0.02517	-0.02790
	SD	0.13102	0.15255	0.13102
	RMSE	0.13389	0.15454	0.13389
				$\alpha^* = 0.90$
200	Bias	-0.02119	-0.02119	-0.02119
	SD	0.10012	0.10012	0.10012
	RMSE	0.10229	0.10229	0.10229
				$\alpha^* = 0.50$

Table 9.10: **Representative results.** AR(1)–ARCH(1): t_5 innovations.

T	Statistic	Whittle	FDEL	Hybrid
75	Bias	-0.04383	-0.02922	-0.04383
	SD	0.15467	0.19249	0.15467
	RMSE	0.16069	0.19460	0.16069
				$\alpha^* = 0.65$
100	Bias	-0.03215	-0.02682	-0.03215
	SD	0.13813	0.15844	0.13813
	RMSE	0.14176	0.16061	0.14176
				$\alpha^* = 0.90$
200	Bias	-0.02204	-0.02113	-0.02204
	SD	0.10985	0.11287	0.10985
	RMSE	0.11198	0.11477	0.11198
				$\alpha^* = 0.60$

Table 9.11: **Representative results.** AR(1)–ARCH(1): $\chi^2(5)$ innovations.

T	Statistic	Whittle	FDEL	Hybrid
75	Bias	-0.04557	-0.03534	-0.04557
	SD	0.14716	0.17190	0.14716
	RMSE	0.15399	0.17542	0.15399
				$\alpha^* = 0.40$
100	Bias	-0.04564	-0.04121	-0.04564
	SD	0.13156	0.14389	0.13156
	RMSE	0.13919	0.14960	0.13919
				$\alpha^* = 0.80$
200	Bias	-0.02853	-0.02793	-0.02853
	SD	0.09715	0.09948	0.09715
	RMSE	0.10121	0.10328	0.10121
				$\alpha^* = 0.65$

Table 9.12: **Representative results.** AR(1)–ARCH(1): Exp(1) innovations.

T	Statistic	Whittle	FDEL	Hybrid
75	Bias	-0.04871	-0.03506	-0.04901
	SD	0.14267	0.17572	0.14183
	RMSE	0.15069	0.17909	**0.15000**
				$\alpha^* = 0.40$
100	Bias	-0.04862	-0.03922	-0.04862
	SD	0.13178	0.15686	0.13178
	RMSE	0.14040	0.16161	0.14040
				$\alpha^* = 0.90$
200	Bias	-0.03302	-0.03247	-0.03302
	SD	0.10271	0.10481	0.10271
	RMSE	0.10784	0.10967	0.10784
				$\alpha^* = 0.90$

AR(1) with Additive Outliers: Additional Results

Table 9.13: **Reprensatative Results.** AR(1)+AO: t_5 innovations.

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.17083	-0.15102	-0.17083
	SD	0.15015	0.19920	0.15015
	RMSE	0.22739	0.24989	0.22739
				$\alpha^* = 0.95$
75	Bias	-0.20891	-0.20643	-0.20891
	SD	0.12997	0.13917	0.12997
	RMSE	0.24601	0.24893	0.24601
				$\alpha^* = 0.95$
100	Bias	-0.19355	-0.19307	-0.19355
	SD	0.11223	0.11416	0.11223
	RMSE	0.22371	0.22426	0.22371
				$\alpha^* = 0.05$
200	Bias	-0.19840	-0.19840	-0.19840
	SD	0.07996	0.07996	0.07996
	RMSE	0.21389	0.21389	0.21389
				$\alpha^* = 0.05$

Table 9.14: **Reprensatative Results.** AR(1)+AO: $\chi^2(5)$ innovations.

T	Statistic	Whittle	FDEL	Hybrid
50	Bias	-0.17142	-0.15481	-0.17142
	SD	0.15650	0.19600	0.15650
	RMSE	0.23206	0.24969	0.23206
				$\alpha^* = 0.45$
75	Bias	-0.21249	-0.20994	-0.21249
	SD	0.12939	0.13886	0.12939
	RMSE	0.24875	0.25167	0.24875
				$\alpha^* = 0.95$
100	Bias	-0.19478	-0.19363	-0.19478
	SD	0.11207	0.11710	0.11207
	RMSE	0.22469	0.22626	0.22469
				$\alpha^* = 0.05$
200	Bias	-0.20485	-0.20485	-0.20485
	SD	0.08329	0.08329	0.08329
	RMSE	0.22112	0.22112	0.22112
				$\alpha^* = 0.05$

Model Misspecification: AR(2) DGM, AR(1) Fit

Table 9.15: **Benchmark results.** Mean spectral distance under AR(2) misspecification: $\mathcal{N}(0,1)$ innovations.

T	Whittle	FDEL	Hybrid
50	0.28725	1.09191	0.28725
			$\alpha^* = 0.05$
75	0.27701	0.97299	0.27701
			$\alpha^* = 0.05$
100	0.27013	0.91231	0.27013
			$\alpha^* = 0.55$
200	0.26498	0.67956	0.26498
			$\alpha^* = 0.45$

Table 9.16: **Representative results.** Mean spectral distance under AR(2) misspecification: $\chi^2(5)$ innovations.

T	Whittle	FDEL	Hybrid
50	0.28908	1.14028	0.28908
			$\alpha^* = 0.50$
75	0.27734	1.03039	0.27734
			$\alpha^* = 0.30$
100	0.27140	0.89646	0.27140
			$\alpha^* = 0.30$
200	0.26609	0.81540	0.26609
			$\alpha^* = 0.45$